

Rheumatology

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Overview

1. Inflammatory Arthritis

- Rheumatoid arthritis
- Seronegative arthritis/Spondyloarthritis

2. Infectious

- Septic
- Gonococcal
- Lyme arthritis

3. Crystal Arthritis

- Gout
- Pseudogout

4. Systemic Autoimmune Rheumatic Disease (SARD)

- SLE
- Sjogren's Syndrome
- Systemic Sclerosis
- Myositis

5. Vasculitis

- Large vessel
- Medium vessel
- Small vessel

6. Inflammatory Disorders

- Still's Disease
- Relapsing Polychondritis

Recent/Expected Guidelines & Updates

- New guidelines published in 2021-2024 :
 - CRA Recommendations for Axial Spondyloarthritis*
 - CRA Updates to RA Management*
 - EULAR/PreS Recommendations for AOSD*
 - EULAR Systemic Sclerosis*
 - EULAR Rheumatic Disease in Pregnancy*
 - ACR/EULAR Antiphospholipid Syndrome Classification Criteria
 - EULAR SLE Guidelines
 - KDIGO Lupus Nephritis Guidelines
 - EULAR ANCA Associated Vasculitis
 - ACR Large Vessel Vasculitis
 - ACR ANCA Associated Vasculitis
 - ACR Vaccination Guidelines
 - ACR Updates to Glucocorticoid-induced Osteoporosis
 - EULAR Cardiovascular Risk in Patients with Rheumatic Disease
 - ACR Peri-Operative Management of Patients with Rheumatic Diseases
 - See PERI-OPERATIVE slides for overview

** in our slides= NEW/UPDATED guidelines*

If anything comes to press and has content relevant for GIM exam we will send you guys an update!

Acronyms used in this talk

Acronym	Full Name	Acronym	Full Name
AAV	ANCA Associated Vasculitis	DMARD	Disease modifying anti-rheumatic drug
MVV	Medium Vessel Vasculitis	MMF	Mycophenolate Mofetil
SVV V	Small Vessel Vasculitis	MTX	Methotrexate
SARD	Systemic Autoimmune Rheumatic Disease	AZA	Azathioprine
IA	Inflammatory Arthritis	HCQ	Hydroxychloroquine
RA	Rheumatoid Arthritis	GC	Glucocorticoids
PsA	Psoriatic Arthritis	SSZ	Sulfasalazine
IIM	Immune Mediated Myositis	csDMARD	Conventional synthetic DMARD (HCQ, MTX, SSZ, LEF)
RPGn	Rapidly Progressive Glomerulonephritis	tsDMARD	Targeted synthetic DMARD (JAK inhibitors)
IBM	Inclusion Body Myositis	bDMARD	Biologic DMARD (TNFi, IL17i, CTLA4, etc...)

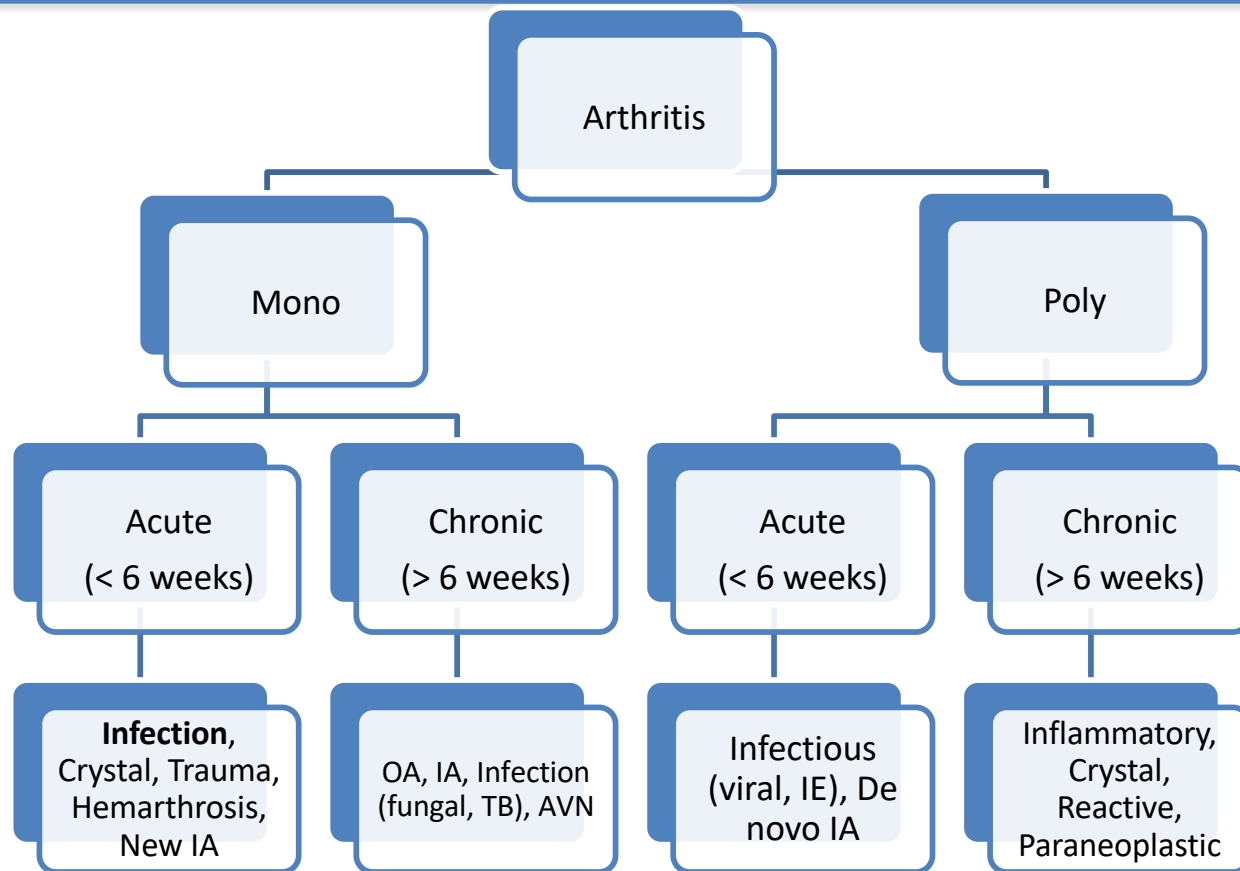
Inflammatory Arthritis

- *Cardinal features of inflammation:*
 - Morning stiffness (typically >60min)
 - Worse with rest, better with activity
 - Night pain (esp. latter half of night)
 - Swelling
- *Joint number:*
 - Mono-arthritis: 1 joint
 - Oligo-arthritis: 2-4 joints
 - Poly-arthritis: 5 or more joints
- *Acute: < 6 weeks duration vs Chronic: > 6 weeks*
- *Extra-articular features*
- *Synovial Fluid Analysis*

Synovial Fluid Analysis

Type of Arthritis	Fluid Appearance	WBC Count (mm ³)	% PMNs	Crystals
Non-inflammatory (OA)	Clear	≤ 2000	50%	No
Inflammatory/ Crystals	Cloudy	>2000-50,000	>50%	- birefring = Gout + birefring = CPPD
Septic	Cloudy/pus	>50,000 bacterial 10k-30k fungal/mycobacterial	>75% PMH indicative of bacterial infection	+/- (Note: crystals can co-exist with septic arthritis - don't be fooled await culture)

Approach to Arthritis

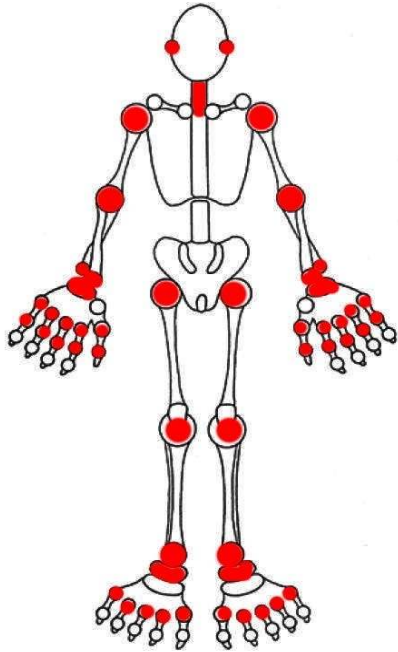


*not an exhaustive list

Chronic Inflammatory Arthritis

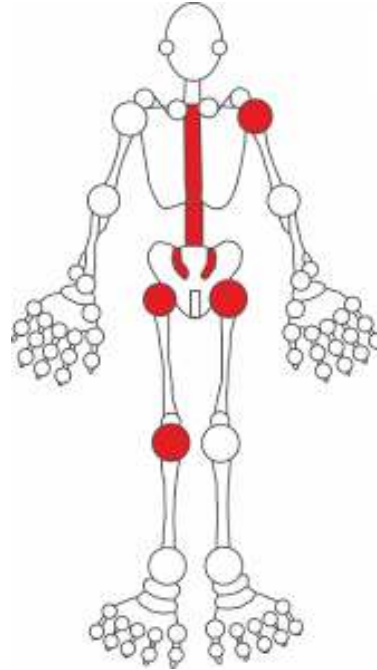
Sero-positive	Sero-negative : “PEAR”
Rheumatoid Arthritis (RA)	Psoriatic Arthritis (PsA)
Systemic Lupus Erythematosus	Enteropathic (IBD) arthritis
Scleroderma (Systemic Sclerosis)	Ankylosing Spondylitis (AS)
Sjogren’s Syndrome	Reactive Arthritis
Dermatomyositis (DM)	Undifferentiated
Polymyositis	
Mixed Connective Tissue Disease	

Patterns of Joint Involvement



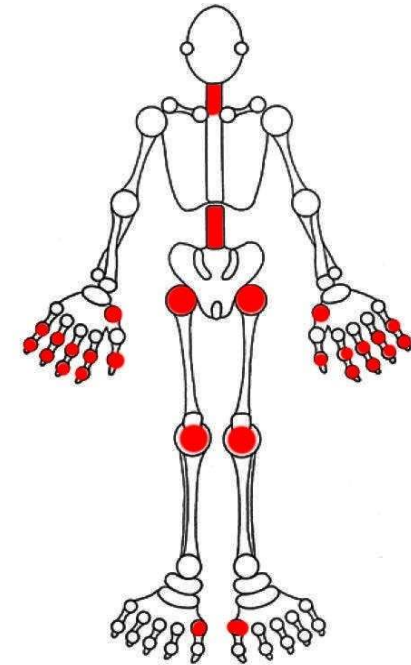
Rheumatoid Arthritis

- Symmetrical
- Wrist, MCPs, PIPs
- MTPs
- DIPs & SPTL spared



Seronegative Spondyloarthropathies (SpA)

- Asymmetric, oligoarticular
- Larger joints, spinal involvement
- Involves DIPs
- Dactylitis, Enthesitis, PsO



Osteoarthritis

- DIP, 1st CMC
- Spine
- Hips, knees
- 1st MTP

MCQ 1 2026



A 56-year-old woman presents with progressive fatigue, arthralgias, and mild hepatomegaly. She reports 30 minutes of morning stiffness affecting the bilateral knees and thumbs. Examination reveals bulkiness of a few joints, but no synovitis, spider angiomas and palmar erythema. Laboratory tests show:

AST 110 U/L, ALT 95 U/L, ALP 220 U/L

Rheumatoid factor 240

Which of the following is the most appropriate next step in management?

- A. Initiate MTX for rheumatoid arthritis
- B. Obtain MSK US to assess for synovitis
- C. Order hepatitis serology
- D. Reassure the patient her RF is non-specific and needs no further follow up

ACR Image Bank

Rheumatoid Arthritis



ACR/EULAR Classification of RA Score ≥ 6 is Definite RA

A) Joint involvement

- 0 = 1 large joint
- 1 = 2-10 large joints
- 2 = 1-3 small joints
- 3 = 4-10 small joints
- 5 = >10 joints, at least 1 small

B) Serology

- 0 = RF, CCP neg
- 2 = RF or CCP low positive
- 3 = HIGH positive ($>3\times$ ULN)

C) Acute Phase Reactants (ESR/CRP)

- 0 = Normal | Abnormal = 1

D) Duration of symptoms

- 0 = $<6\text{wk}$ | 1 = ≥ 6 weeks

Rheumatoid factor differential:

- Other CTD
- Hepatitis C
- Cryoglobulinemia
- Endocarditis
- Malignancy (B cell neoplasms most common)
- PBC
- Syphilis
- Age
- Normal variation

Rheumatoid Arthritis

Classic RA = Symmetric small joint polyarthritis of synovial joints (MCPs, PIPs, wrists)

Do not need serology or X-rays for diagnosis, especially with early disease

- 25% RA are RF negative (RF is not diagnostic)
- Anti-CCP: 95% specificity, can precede arthritis, predicts more erosive disease (in association with smoking = major RF)
- Elevated CRP, ESR may correlate with disease activity

XR findings— periarticular osteopenia, joint space narrowing, marginal erosions

Deformities: swan neck, Boutonniere, ulnar deviation, MCP subluxation



RA Extra-articular Manifestations

Cardiac

- Pericarditis (+/- effusion), myocarditis
- Coronary artery disease (2X risk)
 - RA is independent risk factor for developing CAD due to accelerated atherosclerosis with inflammation

Lung

- Interstitial lung disease (NSIP, UIP)
 - Usually anti-CCP positive
- Pleural effusion (sterile exudate with low glucose)
- Pulmonary nodules
- Bronchiolitis obliterans
- Caplan's syndrome: rheumatoid pneumoconiosis

Hematologic

- Anemia, reactive thrombocytosis
- Felty's syndrome (rare)
 - Seropositive RA + splenomegaly + neutropenia
- Lymphoproliferative conditions
 - May be EBV related or MTX related

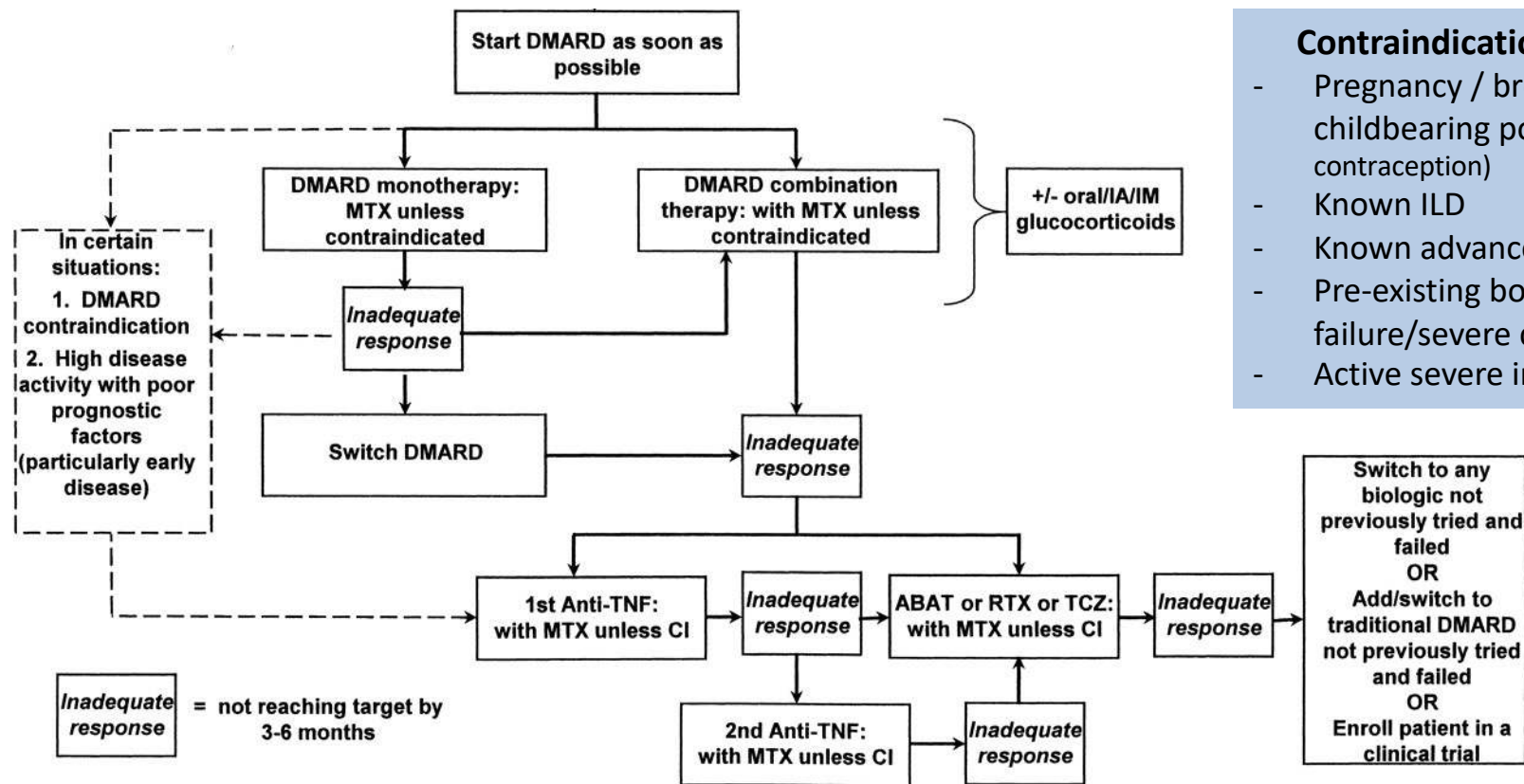
Neurologic

- Carpal tunnel syndrome
 - =One of the earliest signs of RA → clue on oral exam
- C1-C2 instability/subluxation = life-threatening (pre-op oral scenario) 

Other

- Rheumatoid nodules
- Vasculitis – variable vessel involvement
 - Often occurs in “burnt out” disease
- Scleritis/Episcleritis, corneal melt
- Sicca symptoms (dry eyes, dry mouth)
- Raynaud's phenomenon
- Neutrophilic dermatoses
 - (eg) Sweet's Syndrome, Pyoderma gangrenosum

RA Management: Overview



Overview of Therapy

Conventional synthetic DMARD (csDMARD)

- Methotrexate (MTX)
- Hydroxychloroquine (HCQ, PLQ)
- Sulfasalazine (SSZ)
- Leflunomide (LEF)

Biologic DMARD (bDMARD)

- TNF Inhibitors – Adalimumab, infliximab, certolizumab, etanercept, golimumab
- Tocilizumab (IL-6)
- Abatacept (T cell inhibitor)
- Rituximab (anti-CD20)

Small molecules/targeted synthetic DMARD (tsDMARD)

- Tofacitinib* (Xeljanz)
- Baricitinib* (Olmiant)
- Upadacitinib* (Rinvoq)
- Apremilast (Otezla)
 - JAKi – significantly increased risk of herpes zoster

Treat to Target strategy = Treat until patients reach a low disease activity state or remission based on validated disease severity indices

*ORAL Surveillance Study - 2022:

- Large post-marketing safety study revealed increased risk of all-cause mortality, MACE, VTE & cancer in tofacitinib vs TNFi (patients had RA, age ≥ 50 with ≥ 1 CV risk factor)
- **FDA issued black box warning Sep 2021 against all JAK inhibitors for all indications** (approved if failed/intolerant to ≥ 1 anti-TNF)

RELATION Study – Tofa vs TNFi, similar risk of MACE. Tofa not associated with increased risk of cancer excluding NMSC

RA Management

- **“Bridge” Therapy/Symptomatic Treatment** (NOT disease modifying)
 - Steroids (PO, IM, IA) $\leq$ 3 months – *use lowest dose for shortest possible time*
 - NSAIDs, Analgesics
- **Long Term Therapy** (Disease modifying)
 - Step 1: Conventional DMARD
 - **Low disease activity**: Hydroxychloroquine
 - **Moderate to high disease activity**: MTX monotherapy
 - *Note: Triple therapy (MTX + PLQ + SFZ) no longer recommended as a preferred strategy*
 - Step 2: Biologic or Small Molecule
 - Use when failed MTX monotherapy (preferred over triple therapy)
 - Usually start with TNF inhibitor and continue MTX
 - MTX is used for synergistic effects but also to prevent formation of anti-drug Ab
 - Change medication classes to alternate biologic or small molecule if not at target
 - Dose can be reduced in patients with low disease activity or remission for ≥ 6 months

Note: Provincial funding for these drugs may not follow this algorithm. Many require trial of 2+ DMARDs before biologic can be funded.

RA Management: Special Considerations

- **Pulmonary disease**
 - MTX can cause pneumonitis (rare complication)
 - If patient has at baseline parenchymal lung disease that is incidental/mild/stable then can still use MTX over other DMARDs for arthritis particularly if moderate to severe RA disease activity
- **Heart Failure**
 - Don't start TNFi if history of NYHA Class III or IV heart failure
 - If patient develops heart failure on TNFi switch to another agent (non-TNF biologic)
- **Lymphoproliferative Disorder**
 - Use rituximab first line if patient has moderate to severe disease activity and a lymphoproliferative disorder in which rituximab is indicated
- **Hepatitis B**
 - If Hep B core Ab Positive, SAg + → prophylactic antiviral therapy while starting all bDMARD
 - If starting rituximab → prophylactic antiviral therapy if core Ab +
- **Pregnancy**
 - Often goes into remission during pregnancy, risk of post-partum flare
 - NOT all DMARDs safe in pregnancy → must switch to pregnancy safe regimen
- **Metabolic-dysfunction associated steatotic liver disease (formerly NAFLD)**
 - Can still use MTX if liver enzymes normal, liver function tests normal, no advanced liver fibrosis, and patient has moderate to severe disease activity, with more frequent (q4-8wk) monitoring of LFTs

Conventional DMARD Key Points

DMARD	Side Effects	Monitoring
Methotrexate (MTX) <i>(weekly medication)</i>	Hepatotoxicity , nausea, pancytopenia , pulmonary (hypersensitivity pneumonitis, fibrosis), oral ulcers , alopecia, teratogenic Prescribe with folic acid to reduce s/e	-HCV, HBV, CBC, LFTs, Cr +/- CXR -Initial CBC, LFTs, Cr q2-4 weeks, then q12 weeks after 3 months - ESR and CRP for <u>disease</u> monitoring
Hydroxychloroquine	Retinal toxicity , rash, photosensitivity, myotoxicity (rare), cardiotoxicity (rare)	Baseline fundus screening with visual fields plus spectral-domain optical coherence tomography (OCT) AND annually after 5 years of treatment in patients with risk factors. Major RF: duration of use, renal disease, and concomitant tamoxifen.
Leflunomide	GI (nausea, GI pain, dyspepsia, diarrhea), hepatotoxicity , HTN, myelosuppression (rare), peripheral neuropathy, teratogenic	-Baseline CBC, LFTs, Cr, HBV, HCV -Initial CBC, LFTs, Cr q2-4 weeks, then q12 weeks after 3 months
Sulfasalazine	GI toxicity, headache, rash Contraindicated if sulfa allergy	-Initial CBC, LFTs, Cr q2-4 weeks, then q12 weeks after 3 months

2012 CRA Guidelines for management of RA. *J Rheum.* 39(8): 1559-1582.

2015 ACR Guidelines for treatment of RA. *Arth Care Res.* 68(1): 1-25.

2024 CRA Position statement on the safety of Hydroxychloroquine.

Management of low-dose MTX toxicity

Toxicity	Management
Nausea	• Increase folic acid to 5mg daily • Trial H2 blocker or PPI • Add leucovorin post MTX dose
Stomatitis	• Increase folic acid to 5mg daily • Add leucovorin • If folic acid/leucovorin ineffective or ulcers severe: ↓ MTX dose
Hepato-toxicity	• Mildly elevated LFTs= reduce MTX dose • Elevated LFTs > 2x ULN = hold MTX dose then resume at lower dose 1-2 weeks after normalization
Rash	• Dose reduce MTX • If non-resolving, discontinue MTX
Cytopenias	• Dose reduce or discontinue MTX depending on severity
Pneumonitis	• Discontinue MTX, do not re-start

Toxicity with low dose MTX is quite rare, if occurs:

- 1) Folinic acid rescue – reduces MTX bioavailability and replenishes folic acid stores
- 2) Hydration
- help facilitate renal excretion of MTX

Risk factors include: renal impairment, hypoalbuminemia

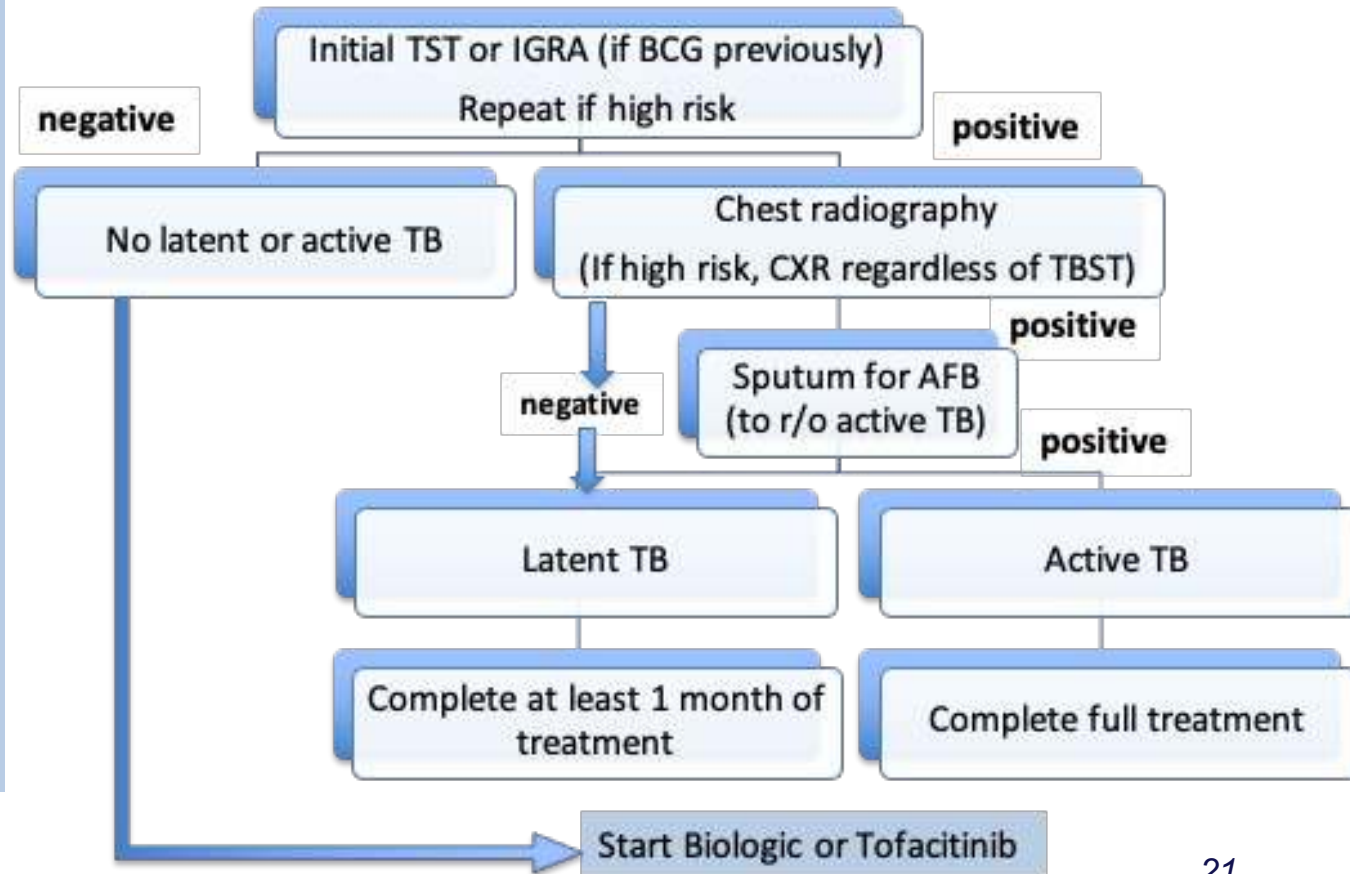
Biologics Key Points

- Risks of Biologics: Infection (new or reactivation), drug induced SLE/antibodies, local skin reactions, malignancy risk
- Baseline Testing (prior to starting biologic): ★
 - Hepatitis B surface Ag, Surface Ab, Core Ab, Hepatitis C serology (*treat concurrently if positive*)
 - TB skin test, IGRA, and/or CXR as appropriate (see algorithm slide)
 - Baselined CBC, LFTs, lipid profile : Tocilizumab, Tofacitinib → repeat lipid profile after 3mo on therapy
- Special Situations/Controversial to use if:
 - NYHA Class III or IV heart failure (**TNFi can worsen CHF**)
 - Active hepatitis (start hepatitis treatment first, consult GI)
 - Prior lymphoproliferative malignancy – use Rituximab
 - Prior solid organ malignancy – consult Oncologist prior to starting
 - Prior skin cancer – csDMARDs preferred (avoid TNFi as increased risk nonmelanoma skin ca)
 - Prior serious infection – considering using csDMARDs if serious infection w/i 12 months
- If flare, modify *frequency* rather than dose first

TB Screening for Biologics or Tofacitinib

HIGH Risk includes:

- Close contacts of persons known/suspected to have active TB
- Born in high incidence country (TB rate of 50/100,000) or travel to country >3 months with high prevalence
 - WHO profiles TB risk by country
https://worldhealthorg.shinyapps.io/tb_profiles/
- Residents/employees of congregate settings (e.g., correctional facilities, and homeless shelters)
- PWID and/or inhaled crack/cocaine
- HIV, immunosuppression



BONUS
Read on own

Vaccination Recommendations

- **Yearly Influenza**
 - High dose recommended for all SARD patients ≥ 65 and patients between 18-65 on immunosuppressants
- **Pneumococcal**
 - Recommended for patients aged < 65 on immunosuppressants
 - PCV13 prime followed by PPSV23 at least 8 weeks later
- **Tetanus toxoid**
 - In accordance with recommendations for general population
- **Hepatitis A and Hepatitis B**
 - Only in patients at high risk of exposure
 - Hep B if non-immune
- **Herpes Zoster**
 - Recommended for patients aged > 18 on immunosuppressants
 - Shingrix (non-live, recombinant) preferred
 - Live attenuated can be administered to high-risk patients ***not on biologics***
 - If starting biologics administer live attenuated ***at least 4 weeks prior*** to initiation
- **HPV**
 - Recommended for patients aged > 26 - < 45 on immunosuppressant and not previously vaccinated
- **COVID 19** – Recommend regular vaccines / boosters per NACI timelines. No drug related contraindications.
- **RSV** - Recommended in Canada to all adults ≥ 75 .

Vaccines should preferably be administered prior to planned immunosuppression, if disease activity permits!

BONUS

Read on own

Vaccination Recommendations

- Medication management around **non-live vaccines**:
 - Methotrexate: Hold 2 weeks after influenza, continue unchanged for all other vaccines
 - May continue depending on disease control
 - Rituximab: Time all vaccines for when next RTX dose is due, then delay RTX for ≥ 2 weeks
 - Prednisone: Give influenza vaccine
 - Defer other vaccines if on ≥ 20 mg Prednisone until tapered
- Medication management around **live-attenuated vaccines**:
 - Live Vaccines = MMR, rotavirus, Zostavax (varicella), chicken pox, flu mist, small pox, yellow fever...
 - Hold treatment for period before:
 - Glucocorticoids, conventional DMARDs (*except IV CYC*): Hold 4 weeks prior
 - Biologics (*except rituximab + IVIG*), IV CYC: Hold 1 prior dosing interval
 - Rituximab: Hold 6 months prior
 - IVIG: Hold for 8-11 months prior (dose dependent – for vaccine efficacy NOT safety)
 - JAKi: Hold 1 week prior
 - ****HOLD ALL Immunosuppressing MEDICATIONS 4 WEEKS AFTER A LIVE VACCINE****
 - Live-attenuated vaccine (rotavirus, BCG) in infants exposed to immunosuppressants in utero
 - Anti-TNF: may receive rotavirus on schedule, BCG delay 6 months
 - Non-TNF, Rituximab: Live vaccines should be avoided within the first 6 months of life

RA Mimics

EORA: elderly onset RA (65y+)

- M=F, usually more drastic initial presentation
- May have “PMR like presentation” – symptoms may be drastically similar early on in presentation
- More larger joint, less likely seropositive

Viral Arthritis

- Often, acute self-limited polyarthritis with viral symptoms
- Parvo B19: often associated reticular rash, childcare worker
- Others: EBV, HBV, HCV



RS3PE: Remitting seronegative symmetrical synovitis with pitting edema

- Acute onset, symmetrical polyarthritis with profound pitting edema
- Most common in older M
- Very responsive to prednisone
- May be paraneoplastic



Hemochromatosis Arthropathy

- Chronic arthropathy, RA-like distribution
- Concurrent CPPD deposition
- Hooked osteophytes at MCPs



MCQ 1 2026



Order hepatitis serologies.

Rheumatoid factor is NOT specific for RA. It may be positive in many chronic inflammatory and infectious conditions RA (remember long RF+ Ddx!). Chronic hepatitis C can be a cause of isolated high titer RF positivity. MTX and HCQ are inappropriate in absence of clear evidence of IA. MTX should be avoided in those with elevated transaminases of underlying liver disease.

A 56-year-old woman presents with progressive fatigue, arthralgias, and mild hepatomegaly. She reports 30 minutes of morning stiffness affecting the bilateral knees and thumbs. Examination reveals bulkiness of a few joints, but no synovitis, spider angiomas and palmar erythema. Laboratory tests show:

AST 110 U/L, ALT 95 U/L, ALP 220 U/L

Rheumatoid factor 240

Which of the following is the most appropriate next step in management?

A. Initiate MTX for rheumatoid arthritis

Seronegative Arthropathies

PsA

Enteropathic (IBD associated)

AS

Reactive Arthritis

- Axial Spondyloarthritis (AxSpA)
 - Primarily affects axial skeleton
 - Sacroiliitis, ankylosis (bony bridging)
- Peripheral Spondyloarthritis
 - Arthritis
 - DIP involvement
- XR Features: ankylosis, periostitis, sacroiliitis (graded 0-4), pencil in cup deformities, erosions



Image from Radiopedia

- SpA Features

- Psoriasis
- Crohn's/colitis
- Dactylitis
- Enthesitis
- Uveitis
- Inflammatory back pain (AWI stiffness, better with PA, responds to NSAIDs, nocturnal waking)
- HLAB27+ (NOT diagnostic)

*Many skin manifestations!
Erythema nodosum, Pyoderma gangrenosum (IBD),
keratoderma blennorrhagicum,
circinate balanitis (Reactive),
Psoriatic skin and nail changes (Psoriasis)



Pitting

Onycholysis

Management of Seronegative Spondyloarthropathies: Axial Disease

Non-pharmacologic: exercise, smoking cessation, physiotherapy, prone laying, vaccines

Pharmacologic:

- **NSAIDs** are first-line therapy; *conditional recommendation* for continuous use > on-demand
- NO steroids or csDMARD for axial disease
- May consider IA steroid injection to SI joints

Step up therapy to BIOLOGICS/SMALL MOLECULE if ongoing high disease activity

Consider presence of any axSpA-associated conditions (i.e. uveitis, psoriasis, IBD) to guide choice

Management of Seronegative Spondyloarthropathies: Axial Disease

- **1st line = TNF- α inhibitors:** Adalimumab, Certolizumab, Golimumab, Etanercept, Infliximab
 - Prefer use of monoclonal TNF if history of IBD or uveitis
 - If fail may switch to another TNF, IL-17 or JAKi
- **1st line = IL-17 inhibitors:** Secukinumab, Ixekizumab, Bimekizumab
 - NO specific recommendation for specific TNF or treatment with an IL-17i over treatment with a TNFi
 - Good option for psoriasis, BUT NOT recommended if have comorbid IBD

If fail a 1st line bDMARD – switch to another bDMARD or tsDMARD (JAKi)

- **JAK inhibitor for axSpA*:** Tofacitinib, Upadacitinib
 - Recommended over no treatment but usually considered/approved if fail bDMARD
 - Conditional recommendation against JAKi as 1st line (TNF/IL-17 preferred)

*May prefer JAKi if concurrent IBD

Patient risks that may cause **caution with JAKi** use – 65+, smoker, significant CV/cancer/VTE risk

Management of Seronegative Spondyloarthropathies: Peripheral Disease

- **Non-pharmacologic:** PT, OT, smoking cessation, weight loss, exercise
- **Pharmacologic**
 - **NSAIDS:**
 - 1st line for peripheral arthritis related to AS or Reactive Arthritis
 - IBD: Prefer Celebrex over other NSAIDs, use with caution as NSAIDs can trigger IBD flare
 - **GLUCOCORTICOIDS:** can consider for peripheral disease, IA for oligo/mono-arthritis
 - Possible risk of paradoxical worsening of psoriasis when tapering prednisone
 - **DMARDS:** Methotrexate, Sulfasalazine, [Leflunomide, Cyclosporine, Apremilast in PsA]
 - **BIOLOGICS/SMALL MOLECULES:**

Class	Drugs	Diseases	Special Considerations
TNF-α	Etanercept, Infliximab, Adalimumab, Certolizumab, Golimumab	AS, IBD, PsA, Reactive	Uveitis/IBD: Infliximab /Adalimumab preferred as most evidence
IL-17	Secukinumab, Ixekizumab, Bimekizumab	AS and PsA	Avoid in IBD
IL-12/23	Ustekinumab	PsA and IBD	Limited role in axial disease
IL-23	Guselkumab, Risankizumab	PsA	Limited role in axial disease
JAKi	Tofacitinib, Upadacitinib	AS, PsA, IBD (UC only)	FDA Black box warning (CV disease, VTE, malignancy, death)
CTLA-4	Abatacept	PsA	Favorable safety profile , especially in terms of infection risk



IA Management in Pregnancy

- **Pre-Pregnancy: Stabilize disease on medication that is safe to take in pregnancy**
 - Discontinue MTX at least 1-3 months prior to conception
 - AVOID leflunomide in patients with the possibility of pregnancy
 - If on leflunomide within 2 years, treat with cholestyramine washout
- **Pregnancy: Maternal disease control is essential. Healthy mom=healthy baby.**
 - Often patients with RA go into remission, seronegative IA runs independent course
 - csDMARD = HCQ, SSZ, AZA, CNI (cyclosporine, tacrolimus) safe to use
 - TNFi = all compatible with pregnancy, Certolizumab marketed as most pregnancy safe
 - Other bDMARD = IL-6/IL-12/IL-23/CTLA4/IL-1 may be used *if needed*. In general, all are of large molecular size thus will have minimal passive placental transfer
 - tsDMARD (JAKi) have very small molecular size → AVOID
 - Avoid NSAIDs after 26 weeks gestation due to risk of premature PDA closure
 - GC can be used, higher doses (>20mg) increase risk of gestational DM, HTN

IA Management in Pregnancy



- **Post-partum: Risk of post partum flare and breastfeeding considerations.**
 - Continue same pregnancy safe regimen, SSZ very rare risk of newborn hemolysis/bloody diarrhea
 - Avoid MTX and leflunomide during breast-feeding
 - Most injectable biologics considered safe as large molecule (most evidence/safety data for TNFi)

Male Preconception

- MTX can be continued
- SSZ possible risk of **reversible** azoospermia
- AVOID CYC (unless pretreatment sperm banking) as irreversible gonadal toxicity

Overview

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- Rheumatoid arthritis
- Seronegative arthritis/Spondyloarthritis

2. Infectious

- Septic
- Gonococcal
- Lyme arthritis

3. Crystal Arthritis

- Gout
- Pseudogout

4. Systemic Autoimmune Rheumatic Disease (SARD)

- SLE
- Sjogren's Syndrome
- Systemic Sclerosis
- Myositis

5. Vasculitis

- Large vessel
- Medium vessel
- Small vessel

6. Inflammatory Disorders

- Still's Disease
- Relapsing Polychondritis

MCQ 2

A 28-year-old man presents with a 6-week history of intermittent right knee swelling and stiffness. He recalls hiking in a wooded area in Connecticut 4 months ago but does not recall a tick bite or rash. He has no fevers, weight loss, or other joint involvement. On exam, there is a right knee effusion, no erythema, no warmth. You perform an arthrocentesis: 30,000 WBCs (predominantly neutrophils), culture negative, crystal negative. Labs: ANA, RF, negative. Lyme ELISA positive with IgG antibodies on western blot.

Which of the following is the most appropriate management?

- A. Oral doxycycline for 28 days
- B. Oral doxycycline for 7 days
- C. IV ceftriaxone for 28 days
- D. Intra-articular corticosteroid injection

Septic Arthritis/Bursitis

- Commonly monoarthritis of hip or knee
- *S. aureus* #1 in both native/prosthetic jts
 - *Salmonella* #1 in osteomyelitis and **septic arthritis in Sickle Cell Disease**
- **Crystal arthritis** can co-exist with infection – don't be fooled!
 - Can also have septic arthritis in patients with known inflammatory arthritis – beware a single joint flare out of keeping with the rest
- Imaging: XR may not see any acute changes, may take 2-3 weeks to develop (periosteal reaction, joint space loss, erosions, subchondral bony destruction)
 - MRI: highest Sn for synovial enhancement, may see joint effusion +/- osteomyelitis
- Empiric antibiotic treatment as per local ID guidelines (e.g. Vanco/Ceftriaxone)
- Definitive management requires source control (i.e. ortho consult for washout)
- Septic bursitis most commonly affects olecranon or pre-patellar bursa
 - RF: DM, baseline bursal disease
 - May develop sympathetic joint effusion from acute inflammatory response/cellulitis

Risk factors: Immunocompromised (alcohol use disorder, DM, CKD, cancer)

- × Prosthetic joints
- × Age (>80y, <5y)
- × Sickle cell (encapsulated organisms)
- × Hypogammaglobulinemia (mycoplasma)
- × Complement deficiencies (Neisseria)

Gonococcal Arthritis

2 common syndromes:

1. Triad of **tenosynovitis**, vesiculopustular **skin lesions**, migratory **polyarthralgias** without purulent arthritis
 2. Purulent arthritis without skin lesions → requires longer course of antibiotics
- Occurs in <5% of gonococcal STDs
 - Treatment:
 - **Ceftriaxone**
 - Check Urine NAAT for Chlamydia and treat if positive (or tx empirically if delays to diagnosis or concern of compliance with follow-up)
 - Doxycycline 100mg po bid x 7 d, or single dose azithromycin 1g



Source: ACR Image Gallery

Lyme Arthritis

See ID Bonus slides for more on Lyme!

- **Clinical Picture**
 - Acute Infection: Arthralgias and myalgias
 - Lyme Arthritis: Late onset (>6 months post infection), oligoarthritis with synovitis and swelling most commonly affecting the knee. Symptoms can fluctuate – swelling and erythema without significant pain. Synovial fluid inflammatory, can see quite high WBC.
- **Diagnosis**
 - Lyme Serology + Clinical Picture
 - Where diagnosis needs confirmation, PCR of synovial fluid or tissue
- **Treatment**
 - **Oral antibiotics x 28 days** (doxycycline or amoxicillin)
 - Treatment failure with objective severe synovitis – ceftriaxone 2-4 weeks IV if other diagnoses excluded
 - Post-antibiotic Lyme Arthritis: Referral to Rheumatology for consideration of DMARD or biologic therapy (repeated courses of IV antibiotics not recommended)

Reactive Arthritis

- Occurs several days to ~4 weeks following **gastroenteritis** or **urethritis**
- Triad of urethritis, conjunctivitis and arthritis (though only seen in 20%)
- Arthritis is typically asymmetric, mono- or **oligoarthritis, lower extremity predominant**
 - Can develop inflammatory back pain and sacroiliitis
- Causative agents: *C. trachomatis*, *Yersinia*, *Salmonella*, *Shigella* & *Campylobacter*
- Eye (50-75%): Uveitis, conjunctivitis
- Reactive arthritis can recur / become chronic (>6mo)
- Treatment:
 - **NSAIDs**, intra-articular corticosteroids
 - Consider DMARDs in recurrent/chronic disease (SSZ preferred choice, variable response other csDMARD, may consider TNF if highly refractory)
 - **NO role for antibiotics** (unless evidence of active infection)

Crystal Arthritis: Gout

Cause: Deposition of **monosodium urate (MSU) crystals** in joints → inflammation

🔪 **Crystals:** Needle-shaped, **Negatively birefringent** under polarized light

💥 **Clinical:** **Acute, self-limited inflammatory arthritis**

– Usually **monoarticular** (classically 1st MTP = podagra)

– Can become **oligo/polyarticular** with recurrences

🔍 **Diagnosis (Gold Standard): Arthrocentesis** → MSU crystals in fluid

⚠️ **Crystals don't exclude infection** → septic arthritis can coexist

RISK FACTORS – “PURINE” mnemonic

P – Purine-rich diet (beer, red meat, seafood)

U – Underexcretion (renal insufficiency)

R – Rx (thiazides, low-dose ASA, pyrazinamide, allopurinol paradox)

I – Increased cell turnover (Tumor lysis, hemolysis, polycythemia)

N – Not enough estrogen (post-menopause → loss of protection)

E – Excess weight (obesity)

Remember: Estrogen = uricosuric → protective → gout rare in young women

Hyperuricemia ≠ gout (but needed for risk, target for therapy)



Source: ACR image gallery

Clinical Pearl:

- Monoarticular in ~80% of initial attacks with predilection to lower extremities, most commonly 1st MTP or knee
- Look for tophi on cool extremities (especially helix/elbows)

Crystal Arthritis: Pseudogout



Calcium pyrophosphate dihydrate disease (CPPD, aka pseudogout)

- Various presentations:
 - Pseudogout –acute mono/oligo-arthritis
 - “RA-like” – chronic inflammatory arthritis
 - OA with CPPD – atypical OA distribution (lateral knees, wrists, elbows)
- Chondrocalcinosis is radiographic diagnosis (may be asx)
- CPPD crystals = rhomboid shaped, positively birefringent
 - If present on synovial fluid, then patient has CPPD
- Secondary Causes: hypothyroidism, hypomagnesemia, hypophosphatemia, hemochromatosis (2nd, 3rd MCP arthritis with hooked osteophytes), hyperparathyroidism, Wilson’s (rare)



Calcific Tendinitis / Milwaukee Shoulder

- basic calcium phosphate/hydroxyapatite crystals
- Older female patients
- Acute onset, destructive shoulder arthropathy

Crowned Dens Syndrome: acute or subacute onset upper neck pain (usually with limited ROM), elevated inflammatory markers and often fever
Diagnostic of CPPD if clinical/imaging features of CDS present

Treatment of Gout: Acute★

NSAIDs

Colchicine

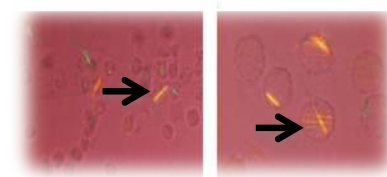
- 1.2mg load then 0.6mg an hour later then 0.6mg BID until symptoms resolve
 - Requires dose reduction for CKD – CrCl <30 start 0.3mg per day, avoid in CrCl <10 or on dialysis
- Risks: diarrhea, nausea, vomiting, (myopathy for long term use)
 - Overdose is life-threatening and is non dialyzable with no antidote

Glucocorticoids (IA / PO)

- IA if mono/oligo-arthritis, oral if polyarthritis (or C/I to above meds)

IL-1 blocker (Anakinra)

- Consider only in patients with frequent flares and contraindications to colchicine, NSAIDs & corticosteroids



Frequently asked Allopurinol/ULT Facts:

- If patient is already on chronic allopurinol/ULT therapy you do NOT need to stop it during an acute flare
- Common concern is ↑ in gout attacks during initiation of ULT thus must add prophylaxis (e.g colchicine while starting)
- Do NOT use ULT with azathioprine = risk of bone marrow failure
- If develop allopurinol hypersensitivity reaction (rash, fever, ↓ platelets, ↑ liver enzymes) STOP and never take again

Treatment of Gout: Chronic

Non-pharmacologic: dietary changes, alcohol reduction, med review

Pharmacologic: Urate lowering therapy

Definite indications:

- Two or more attacks/yr
- Tophaceous gout
- Gouty arthropathy (erosions)

Conditional indications: 1 episode acute gout + RF:

- CKD Stage 3+
- Uric acid level > 535
- Urolithiasis

Allopurinol vs. Febuxostat* - some early studies suggested increased CV events w febuxostat ; FDA issued black box warning re CV safety. Since then - FAST trial (Lancet, 2020) → “Febuxostat is non-inferior to allopurinol therapy with respect to the primary cardiovascular endpoint, and its long-term use is not associated with an increased risk of death or serious adverse events compared with allopurinol.”

ULT Treatment Options:

1) Allopurinol: 1st line

- Start 100mg/d (or 50mg if CKD 4) and up-titrate until reach target uric acid level < 356umol/L or <300umol/L if tophi
- Can start in addition to colchicine during flare
- Continue during flare if already on treatment at baseline
 - Risk of hypersensitivity syndrome: TEN/SJS, fever, eosinophilia, hepatic necrosis, nephritis, diarrhea
 - Consider testing HLA-B*5801 in Southeast Asian and black patients who are at increased risk

2) Febuxostat (\$\$)

- Use if cannot tolerate allopurinol or high risk for allopurinol hypersensitivity, not approved for dialysis patients
- Risk of rash, diarrhea

Overlap with anti-inflammatory prophylaxis with either colchicine, NSAID or low-dose GC for 3-6 months

MCQ 2

A 28-year-old man presents with a 6-week history of joint swelling and stiffness. He recalls hiking in a wooded area 2 weeks ago but does not recall a tick bite or rash. He has no other symptoms. Joint involvement. On exam, there is a right knee joint with warmth. You perform an arthrocentesis: synovial fluid (many neutrophils), culture negative, crystal negative. Serology: ELISA positive with IgG antibodies on week 6.

Which of the following is the most appropriate treatment?

- A. Oral doxycycline for 28 days
- B. Oral doxycycline for 7 days
- C. IV ceftriaxone for 28 days
- D. Intra-articular corticosteroid injection

Lyme arthritis is a late manifestation of *Borrelia burgdorferi* infection occurring 6mo-2Y post initial infection. Diagnosis can be made with clinical features and serum antibody testing over synovial fluid culture. Presents usually oligoarticular, can be migratory and most often affects the knee. **First line treatment is with doxycycline for 28 days** (strong recommendation), thus 7 days is insufficient. IV ceftriaxone is reserved for those with refractory arthritis. IA can be used for symptom management, but would await final joint culture first AND the most definitive treatment is doxycycline. If fail 1 oral and 1 IV antibiotics, may consider DMARD. Antibiotic therapy longer than 8 weeks is unlikely to confer any additional benefit.

Overview

1. Inflammatory Arthritis

- Rheumatoid arthritis
- Seronegative arthritis/Spondyloarthritis

2. Infectious

- Septic
- Gonococcal
- Lyme arthritis

3. Crystal Arthritis

- Gout
- Pseudogout

4. Systemic Autoimmune Rheumatic Disease (SARD)

- SLE
- Sjogren's Syndrome
- Systemic Sclerosis
- Myositis

5. Vasculitis

- Large vessel
- Medium vessel
- Small vessel

6. Inflammatory Disorders

- Still's Disease
- Relapsing Polychondritis

MCQ 3

A 33 year old female presents to ED with 3 weeks of fever, general malaise, progressive fatigue, arthralgias, and cough. Her vitals are as follows: T 39.4, HR 100, BP 90/65mmHg, SpO2 91%. Her labs are as follows: Hb 100, WBC 3.5, Plts 120, Cr 75, UA benign, ANA 1:160, dsDNA 7, C3 0.11, C4 1.0. On your assessment, she appears unwell. She has scarring alopecia and a malar rash on your assessment.

Which of the following is the most appropriate next step in management?

- A. IV methylprednisolone 500 mg
- B. High dose oral steroids
- C. Hydroxychloroquine
- D. IV antibiotics

2019 SLE Classification Criteria

Entry Criterion: ANA titre $\geq 1:80$

If **absent**, do not
classify as SLE

If **present**, apply
additive criteria

96.1% sensitivity
93.4% specificity

Weighted score
based classification
requiring **1 clinical
criteria + ≥ 10
points**

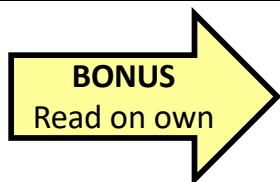
2019 SLE Classification Criteria

Clinical domains and criteria	Wt	Immunology domain/criteria	Wt
Fever	2	Antiphospholipid antibodies	
Hematologic			
Leukopenia	3	Anti-cardiolipin antibodies OR	2
Thrombocytopenia	4	Anti-B2GP1 antibodies OR	
Autoimmune hemolysis	4	Lupus anticoagulant	
Neuropsychiatric			
Delirium	2		
Psychosis	3		
Seizure	5		
Mucocutaneous		Complement proteins	
Non-scarring alopecia	2	Low C3 OR low C4	3
Oral ulcers	2	Low C3 AND low C4	4
Subacute cutaneous or discoid lupus	4		
Acute cutaneous lupus	6		
Serosal		SLE-specific antibodies	
Pleural or pericardial effusion	5	Anti-dsDNA antibody* OR	6
Acute pericarditis	6	Anti-Smith antibody	
Musculoskeletal : Joint involvement	6		
Renal			
Proteinuria >0.5g/24h	4		
Renal bx class II or V lupus nephritis	8		
Renal bx class III or IV lupus nephritis	10		

Cutoffs for Criteria:

- Fever >38.3°C
- Leukopenia WBC <4.0
- Thrombocytopenia <100
- Pericarditis - ≥ 2 of: rub, typical pleuritic chest pain, ECG ST elevation w/ PR depression, new/worsening effusion on imaging
- Joint involvement = synovitis in 2 or more joints or tenderness in 2 or more joints w/ >30 min am stiffness

2019 ACR/EULAR SLE Classification
Criteria. *Arth Rheum.* 71(9):1151-1159. 46



Malar Rash vs Rosacea



Characteristics of **malar rash**:

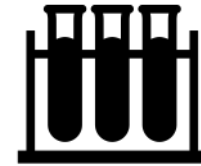
- Lasts a few days to weeks
- Spares nasolabial folds
- Precipitated by sun exposure
- Look for other systemic manifestations!



Characteristics of **rosacea**:

- Frequent flushing (lasting few hours)
- Telangectasias, papules, pustules
- Aggravated by sun, spicy foods/drinks, EtOH
- Can cross nasolabial fold

Lupus Labs



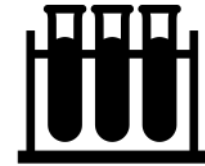
ANA: ANA $\geq$ 1:80 is entry criterion

- Sensitive (95%), but not specific
- DDx of positive ANA
 - Rheum: SLE, scleroderma, MCTD, drug-induced lupus, polymyositis/dermatomyositis, rheumatoid arthritis
 - Thyroid disease (ex Hashimoto's), autoimmune hepatitis, PBC, IBD, IPF
 - Infection: hepatitis C, parvovirus, TB
 - Family history of any of above
 - Healthy (healthy titres: 1/40 = 20%, 1/80 = 10%, >1/160 = 5%)
- Not used to follow disease activity (do not repeat over time)

Anti-dsDNA

- Specific (97-98%), but not sensitive (present in 60-80% of SLE patients)
- Associated with lupus nephritis
- Can be used for monitoring disease activity, along with serum complements (C3, C4) in concordant individuals
 - Concordance = When flare, high anti-dsDNA, low complements
 - Discordance = When flare, above pattern doesn't apply

Lupus Labs: ENA



- Anti-Sm: highly specific but not sensitive (30-40%)
- Anti-histone: **Drug-induced lupus**; SLE (50-70%)
- Anti-RNP: **Required for diagnosis of MCTD**; SLE (30-40%)
- Anti-Ro (SSA) / Anti-La (SSB): Sjogren's syndrome; risk of **congenital heart block and neonatal cutaneous lupus** even if mother has no SARD

Drug induced lupus: (classically – ANA+, Anti-histone+, dsDNA –ve)

#1 causative drug → Procainamide. Others – Hydralazine, Isoniazid, TNFi - *Anti-TNF-induced lupus (ATIL)* – see much more +dsDNA. Rarely: phenytoin, carbamazepine

Symptoms often develop after months to years of exposure to the drug

STOP the offending drug & symptoms will resolve. Reversibility is the key that it is NOT true SLE. Note ANA+ may persist >1y after drug is stopped.

Overview of SLE Management

HCQ (hydroxychloroquine) should be background therapy for **ALL** SLE patients

HCQ = ↑ survival & ↓ risk of renal flare

- Any additional therapy is based on organ involvement and severity!

General/Non-pharmacologic Measures:

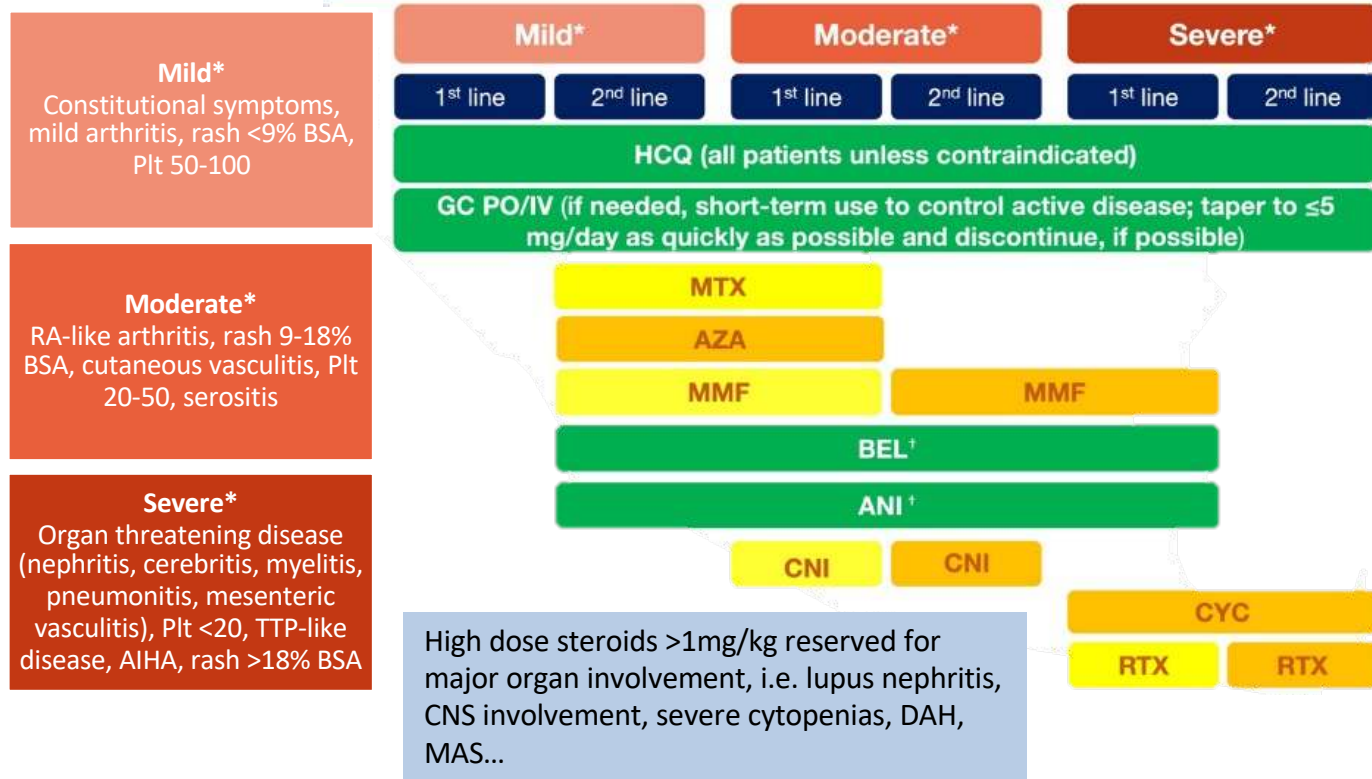
- Avoid triggers – sunlight, infections, smoking, sulfa drugs, echinacea, alfalfa sprouts
- Sun protection
- Regular CV risk assessment – mitigate risk via BP control, normal weight, lipid + glucose control
- Smoking cessation – may interfere with effectiveness of anti-malarials and certain biologics
- Bone protection
- UTD vaccinations – HZ, HPV, COVID, influenza, Pneumococcal
- Preventative health care – physical activity, cancer screening
 - CRA: cervical cancer screening in SLE patients on ANNUAL basis regardless of immunosuppression

Lupus Nephritis General Measures:

- RAAS blockage
- Consider SGLT2i if reduced eGFR
- Consider VKA/heparin if concomitant APS nephropathy

Management of Non-Renal SLE

Treatment of Non-Renal Systemic Lupus Erythematosus



Biologics for SLE

ANI: Anifrolumab (anti-INF 1)

BEL: Belimumab (anti-B cell activating factor)

BOTH control disease activity, reduce flares and reduce GC dose

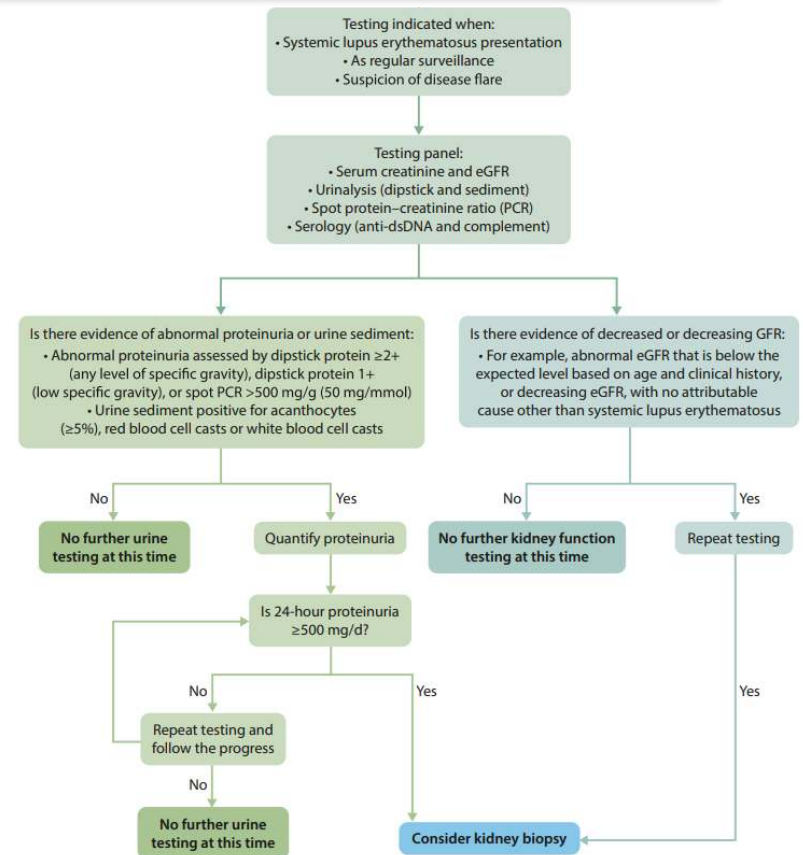
Recommendation of belimumab and anifrolumab 1st line in severe disease refers to cases of extrarenal SLE with non-major organ involvement, but extensive disease from skin, joints

Use of anifrolumab – great for cutaneous involvement

ANI/BEL **NOT** recommended for patients with neuropsychiatric disease

Lupus Nephritis: Overview

- Biopsy ALL SLE with:
 - Glomerular hematuria and/or cellular casts
 - Proteinuria $>0.5\text{g}/24\text{h}$ or spot UPCR $>500\text{ mg/g}$
 - Unexplained $\downarrow$ in GFR
- Treat with HCQ unless contraindications
- Avoid high sodium diet
- Optimize BP with goal $<130/80\text{mmHg}$ (EULAR)
- Renoprotective medications – RAAS blockage, SGLT2i
- Avoid nephrotoxic insults/medications
- VKA/heparin if concurrent APS nephropathy



Lupus Nephritis: Overview of management

Overview of Management HCQ + RAASi for all classes

Class I: Minimal Mesangial LN

Class II: Mesangial Proliferative LN

Class III: Focal LN (<50% of glomeruli)

Class IV: Diffuse LN ($\geq$ 50% of glomeruli)

Class V: Membranous LN

Class VI: Advanced sclerotic LN (90% of glomeruli globally sclerosed w/o residual activity)

Other Lupus Renal Syndrome— SLE + TMA :
Different causes! TTP, SLE-APS, Complement mediated

→ Treatment guided by extra-renal disease unless nephrotic syndrome then will consider GC +/- immunosuppression

→ Aggressive immunosuppression if class III/IV or class V with >1g proteinuria
Anti-proteinuric/anti-hypertensive agents

→ Supportive Therapy
Treat extra-renal manifestations

→ Measure ADAMSTS if you suspect SLE-TMA
Measure APLAS
Treat underlying cause (PLEX, steroids, anticoag, eculizumab, etc)

Lupus Nephritis: Class III-V



- HCQ + RAAS for ALL
 - Up front triple therapy
 - GC + MMF + Belimumab (BEL)/
Calcineurin inhibitors (CNI)
 - BEL preferred if extra-renal manifestations
 - CNI preferred if proteinuria >3g/d
- **Class 5 now also treated this way if proteinuria >1g/d**
- If <1g/d then GC +/- immunosuppression

Lupus Nephritis

2023 EULAR guidelines, 2024 KDIGO Guidelines, ACR 2024

*ACR 2024: Lower steroid use recommended
IV MP 250mg-1000mg for 1-3 days followed
by lower oral GC 0.5mg/kg/d (max dose
40mg) with goal of 5mg by 6 mo

Class III/Class IV/Class V with >1g protein: May present with hematuria, proteinuria, hypertension, ARF
Induction: High dose glucocorticoids* + another agent (MMF alone or combo with BEL/CNI OR CYC)

Mycophenolate mofetil (MMF):

- Conditionally recommended over CYC, especially for pts with desire for future fertility
- Side effects: GI upset, cytopenias, unsafe in pregnancy

Cyclophosphamide (CYC):

- Low-dose IV protocol (500 mg q2 weeks x 6) preferred
- Risks infertility (significant, esp ♂), infection, GU malignancies, hemorrhagic cystitis, cytopenias
- Used less often due to side effects, if used after induction will be changed to MMF or alternative

CNI (calcineurin inhibitors i.e. voclosporin, tacrolimus)

- Prefer if high level of proteinuria
- Side effects: hypertension, AKI

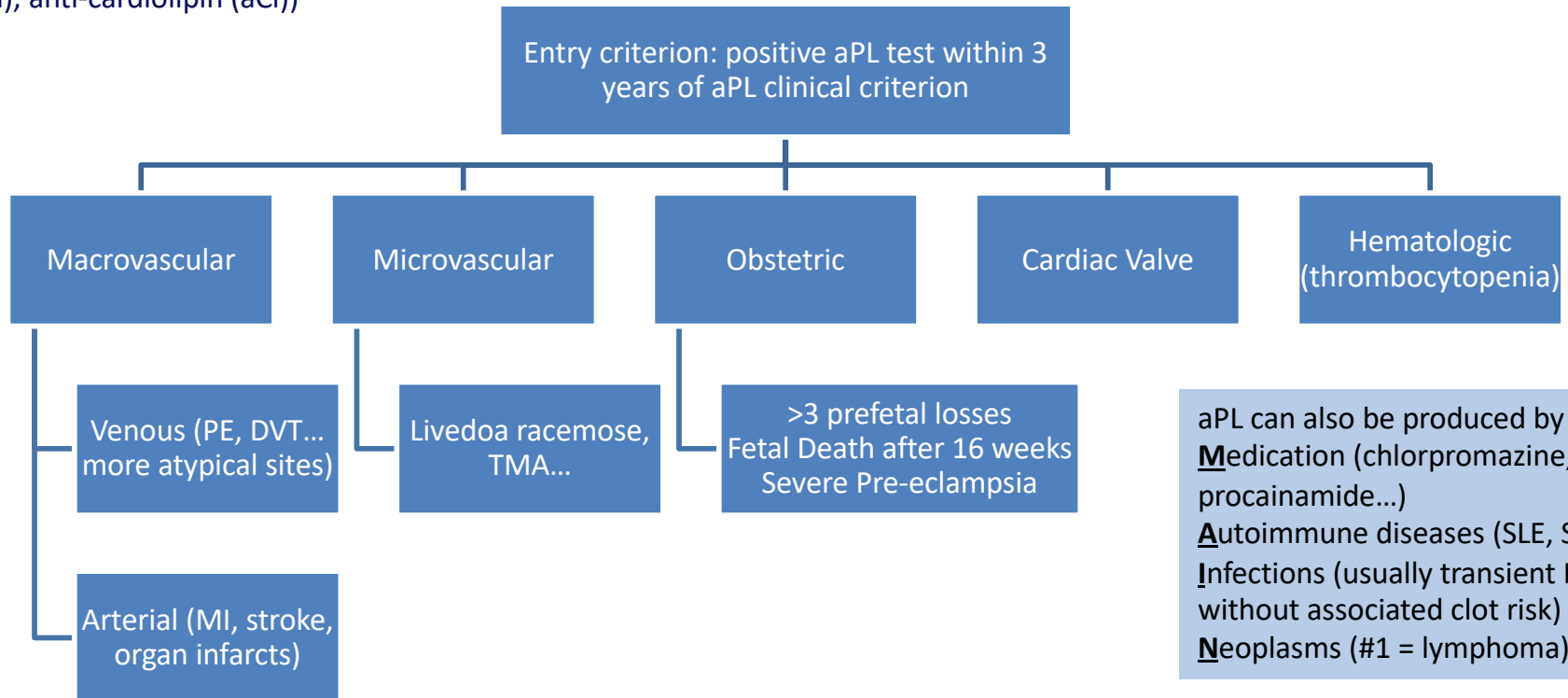
Belimumab (IV or SC)

- Improved renal response, prefer if extra-renal disease

Treatment of Lupus Nephritis	
Initial	Subsequent
HCQ (all patients unless contraindicated)	
GC PO/IV (consider pulse IV MP, then 0.3-0.5 mg/kg/day depending on severity; taper to ≤ 5 mg/day as quickly as possible)	
MMF	
Low-dose CYC	AZA/MMF
MMF/low-dose CYC + BEL [§]	MMF/AZA + BEL [§]
MMF + CNI (esp. VOC, TAC) [^]	
High-dose CYC ^{*,¶}	
RTX [†]	
Any of the above-mentioned unless contraindicated [^]	

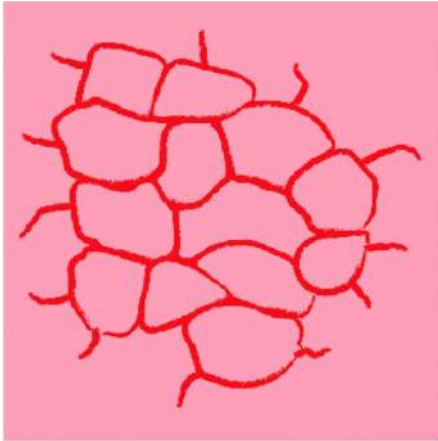
Anti-Phospholipid Syndrome (APS)

Systemic autoimmune disease with arterial, venous or microvascular thrombosis, pregnancy morbidity or non-thrombotic manifestations in patients with persistent antiphospholipid (aPL) antibodies (lupus anticoagulant (LAC), anti-B2-glycoprotein (anti-B2-GPI), anti-cardiolipin (aCI))

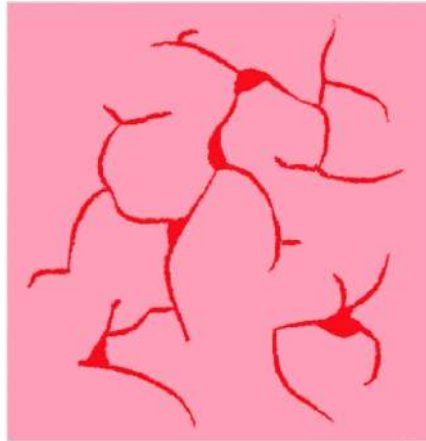


BONUS
Read on own

Livedo Racemosa & Vasculopathy



Livedo reticularis



Livedo racemosa

Livedo racemosa – violaceous net like.
Blotching of skin – broken / nonuniform discoloration (vs livedo reticularis)



Livedo vasculopathy – otherwise unexplained painful papules or purpuric plaques → biopsy showing thrombosis

Clinical Domains	Wt	Laboratory Domain	Wt
Macrovascular (Venous)		Lupus Anticoagulant	
VTE with high risk VTE profile*	1		
VTE without high risk VTE profile	3	One time LAC positive	1
		Persistent LAC positive	5
Macrovascular (Arterial)		Note: persistent aPL – 2 measures >12 weeks apart	
AT with high risk CVD profile**	2		
AT without high risk CVD profile	4		
Microvascular			
Suspected	2		
Established	5		
Obstetric		aCL or Anti-B2- GPI	
≥3 consecutive pre-fetal (<10w) and/or early fetal deaths (10-15+6)	1	Moderate or high positive IgM aCL +/- aB2-GPI)	1
Fetal death in absence of PEC/PI with severe features	1	Moderate positive IgG	4
PEC OR PI with severe features <34w with or w/o fetal death	3	High positive IgG aCL OR aB2-GPI	5
PEC AND PI with severe features <34w w or w/o fetal death	4	High positive IgG aCL AND aB2-GPI	7
		PEC: pre-eclampsia, PI: placental insufficiency	
Cardiac Valve		Note: Moderate (40–79 units) and high (>80 units) for aCL and aB2-GPI on ELISA	
Thickening	2		
Vegetation	4		
Hematology : Thrombocytopenia	2		
Plt nadir 20-130x10 ⁹ /L			

Entry Criterion: 1 documented clinical criterion PLUS positive aPL within 3 years of clinical criterion

If **absent**, do not classify as APS

If **present**, apply additive criteria

Weighted score based classification requiring **3 clinical domain points AND + 3 lab domain points**

***VTE High Risk Profile:** if a patient developed a VTE after a surgery or in context of malignancy this would be considered high risk as there is a clear provoking event thus this would be scored lower than an individual who has an unprovoked VTE (esp if an atypical site, i.e. pelvic, ocular...)

****High Risk CVD Profile:** same applies here, a 69M with DM, CKD, HTN who has an MI is considered high risk CVD profile and scored less than a 32F with no clear CV risk factors who has an MI

BONUS
Read on own

Catastrophic Antiphospholipid Syndrome (CAPS)

- Widespread coagulopathy associated with aPL
- >3 or more organs involved (predominantly small vessel vascular occlusive events) occurring within 7 days with positive aPL
 - May see renal involvement (ARF, hematuria), lung (ARDS, PE, DAH), CNS (stroke, seizures), skin (livedous changes), heme (AIHA, thrombocytopenia)
- Develop multi-organ failure from thrombotic storm
- More microvascular, arterial events than in regular APS
- Quite rare, but can be initial APS presentation in >50% of patients

Ddx: CAPS

- Sepsis
- DIC
- TMA
- HIT
- HELLP

APS Management

EULAR 2023 SLE guidelines



- Thrombotic APS
 - Vitamin K antagonist therapy for life (warfarin preferred agent)
 - DOAC use is associated with ↑ risk of arterial/venous thrombotic events [meta-analysis]
- High risk APS profile without prior thrombosis
 - High risk = LAC + or double or triple positivity
 - ASA 81 mg “for life”
- Obstetric APS
 - ASA + **prophylactic** dosing LMWH during pregnancy and 6 weeks post-partum
- Catastrophic APS
 - Full dose anticoagulation (UFH)
 - High dose glucocorticoids + PLEX or IVIG
 - Refractory: eculizumab or rituximab or CYC may be considered

Lupus and Pregnancy



- Avoid pregnancy until disease is well controlled!
 - Low disease activity *prior* to pregnancy leads to improved maternal and fetal outcomes
- Steroids, HCQ, AZA, CNI, Belimumab are considered safe immunosuppressive treatments during pregnancy
 - Must demonstrate that disease remains stable on pregnancy safe regimen prior to conception attempts
- Start ASA 162mg QHS prior to 16 weeks gestation to reduce pre-eclampsia risk

Special Considerations

Lupus Nephritis

- Ensure LN is inactive for >6mo, proteinuria well controlled, BP well controlled
- Must stop ACEi when pregnant

Ro/La+

- Risk of congenital heart block and neonatal SLE (transient disease related to maternal Ab)
- HCQ for all even if no history of SARD
- Refer for fetal echo week 16-26 (frequency depends on antibody titer and if prior pregnancy affected)
- Positive aPL:
- No APS= ASA alone
- OB APS= ASA 81mg + prophylactic heparin until 6-12 weeks PP
- Thrombotic APS= ASA 81mg + therapeutic heparin during pregnancy and PP

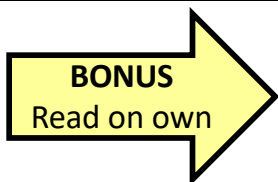
MCQ 3

A 33 year old female presents to ED with 3 weeks of fever, general malaise, progressive fatigue, arthralgias, and cough. Her vitals are as follows: T 38.4, HR 100, BP 90/65, RR 16, SpO2 94% on RA. Her labs are as follows: Hb 100, WBC 18.5, ESR 45, CRP 10, ANA 1:1000, dsDNA 1:100, C4 1.0. On your assessment, she appears to have a systemic illness. What is the most appropriate next step in your assessment.

Which of the following is the most appropriate next step in your assessment?

- A. IV methylprednisolone 500 mg
- B. High dose oral steroids
- C. Hydroxychloroquine
- D. **IV antibiotics**

IV antibiotics. This patient does not have a known diagnosis of SLE. She may in fact have underlying SLE, but the most appropriate first step in treatment is to empirically cover for possible infection. Infections can be a trigger for SLE flares, but there will be no benefit to acute initiation of hydroxychloroquine. There is no clear major organ involvement at present time to warrant initiation of high dose steroids (IV or oral).



High Yield Lupus Trivia

Drug Induced Lupus:

- Common: hydralazine, procainamide, quinidine
- Less common but still implicated: isoniazid, PTU, minocycline, Carbamazepine, penicillamine, methyldopa, sulfasalazine, **TNF inhibitors**
 - For TNFi: often get baseline ANA, anti-dsDNA so can compare if symptoms develop on tx
- Serology: ANA+, dsDNA -, anti-Sm -, **anti-histone Ab+** (90-95%, vs 60% for idiopathic SLE)
- TNFi usually anti-histone -ve; dsDNA+.
- Treatment: discontinue offending med, +NSAID, other treatment rarely needed (consider topical steroid if cutaneous, HCQ if derm/msk, rarely po steroids)

Shrinking Lung Syndrome: rare complication related to diaphragmatic muscle weakness

- Lungs are clear on CT imaging but volume is decreased (MIP/MEPs), reduced volumes on PFTS
- *Optimal treatment is unknown – super rare manifestation of SLE*

Libman Sacks Endocarditis (Non-infectious endocarditis): associated with APLAS

- Thrombus on valve consists of accumulations of immune complexes, mononuclear cells, hematoxylin bodies, and fibrin and platelet thrombi
- Can result in embolic phenomena
- Treated with steroids and anticoagulation

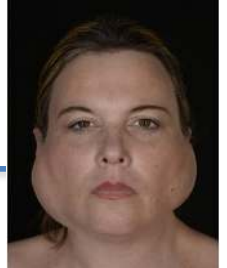
MCQ 4

A 45-year-old woman presents with progressive shortness of breath on exertion and one year history of Raynaud's phenomenon. She reports skin tightening of the hands and face over the past 2 years and difficulty swallowing solids. On examination, you note puffy digits and skin thickening extending to the wrists with few digital tip ulcers. You note bibasilar fine crackles on lung auscultation. Investigations: ANA 1:320, Anti-Scl-70 >8. Pulmonary function tests show FVC 65% predicted, DLCO 45% predicted. Echocardiogram shows normal biventricular function with RVSP of 30mmHg.

Which of the following is the most likely cause of her pulmonary findings?

- A. Pulmonary arterial hypertension
- B. Interstitial lung disease
- C. Left-sided heart failure
- D. Chronic aspiration pneumonitis

Sjogren's Syndrome



Source: ACR Image Gallery

ACR/EULAR 2016 Criteria: 1 symptom of ocular or oral dryness with score >4

- Minor alivary gland biopsy with focus score >1 ("lip biopsy") → focal lymphocytic sialadenitis
- Anti-SSA positive
- Schirmer test <5mm/5m in at least 1 eye
- Unstimulated salivary flow rate <0.1ml/min
- Ocular stain score >5

Clinical manifestations:

- Mild: Xerostomia, keratoconjunctivitis
- Severe: florid salivary gland enlargement, adenopathy, extra-glandular sx
- Non glandular involvement: Arthritis, **vasculitis**, demyelinating neuropathy, glomerulonephritis, tubulointerstitial nephritis causing renal tubular acidosis
- Serology: ANA, **Anti-Ro**, **Anti-La**, RF

Epidemiology:

- **>40X increased risk of B-cell lymphoma**
- Can be secondary to other autoimmune diseases (i.e. RA, SLE...)

Ddx: bilateral parotid gland enlargement

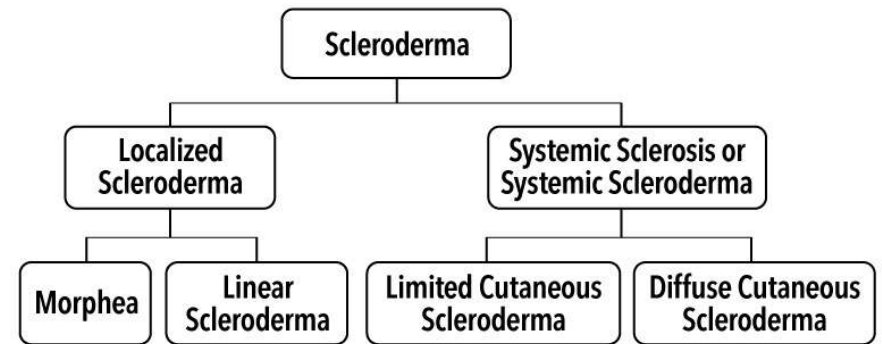
- Sjogren's
- Infectious – Mumps, TB, bacterial , Hep C, HIV
- Sarcoidosis
- Lymphoma
- Alcoholism, Anorexia/bulimia
- IgG4 related disease*

*Sending IgG4 is now routine – esp if enlargement of submandibular glands / lacrimal glands **without parotid enlargement**.

Beware –if lacrimal gland enlargement even in the context of Sjogren's syndrome, **raises concern for lymphoma**

Systemic Sclerosis

- Localized:
 - Skin limited disease *without* internal organ involvement
- Limited:
 - Skin thickening **DISTAL** to the elbows/knees
- Diffuse:
 - Skin thickening beyond elbows/knees



Sclerodactyly

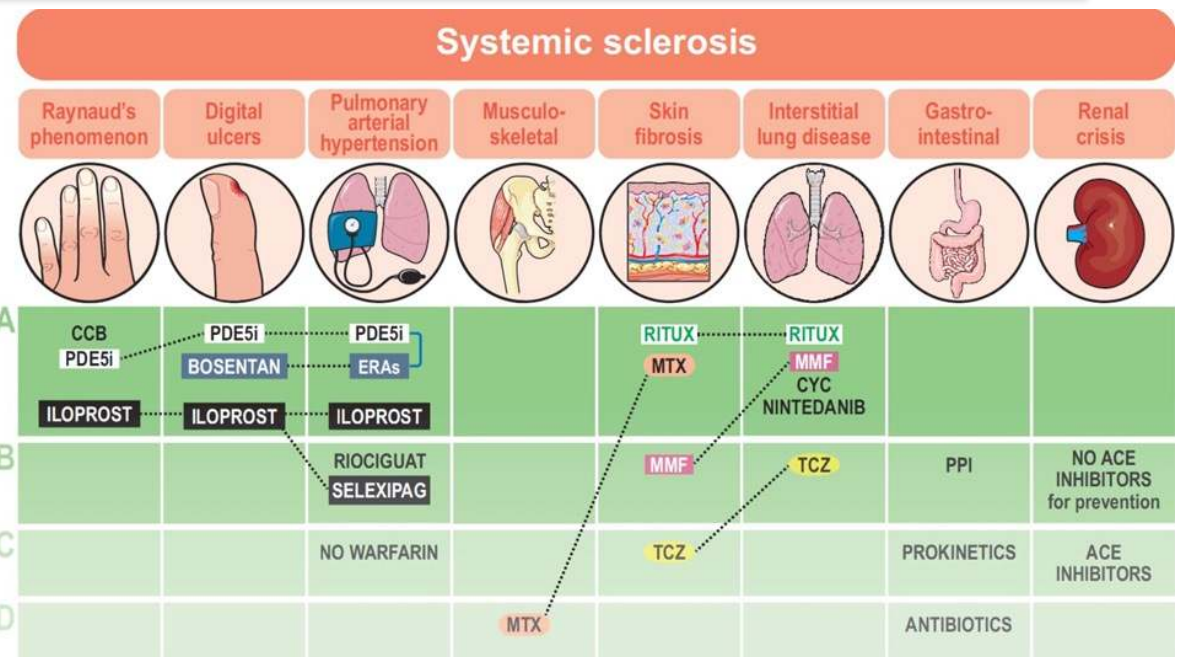


Calcinosis

Source: Dr McKenna's office

Systemic Sclerosis

- Serology: highly specific for scleroderma >98%
 - Anti-centromere: **Limited 60%**, Diffuse 15%, screen for primary biliary cirrhosis
 - Anti Scl-70/topoisomerase I: ~40% of scleroderma patients, mostly **diffuse** disease
 - Anti-RNAPolIII: risk of SRC
- Diffuse:
 - Worse prognosis as more abrupt onset (RP within 1yr or developin other features)
 - Higher risk of ILD and SRC
- Limited:
 - Usually presents after longstanding RP



"CREST" Syndrome – outdated as all of these can occur in both limited and diffuse forms

Calcinosis / Raynaud's phenomenon / Esophageal dysfunction
Sclerodactyly / Telangiectasias

Complications of Systemic Sclerosis

- **Scleroderma Renal Crisis** ★

- 10-20% of **diffuse** systemic sclerosis
- **Increased risk with prednisone**, RNAPol3 autoantibodies, early diffuse dx
- Acute/progressive renal failure + HTN + usually bland UA
- Can have normotensive SRC (rare)
- Treatment: **ACE inhibitor (Captopril)**

SSc 5 main manifestations:

1. Skin
2. Raynaud's +/- ulcers
3. Lung (ILD, PH)
4. GI
5. Kidney (SRC)

- **Pulmonary Hypertension**

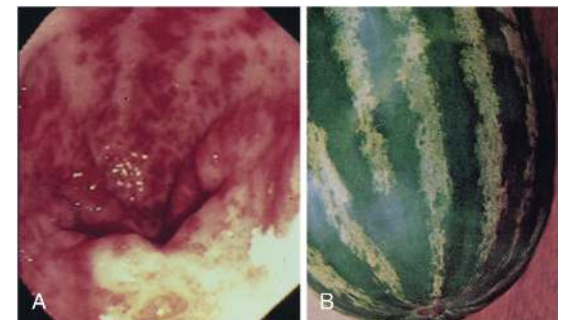
- Group 1 most common, occurs 5-19% of all systemic sclerosis; more common in **limited**
- Monitor with NT-proBNP, echo and PFTs for all SSc patients annually

- **Interstitial Lung Disease**

- Screening with PFTs and HRCT at baseline → especially for HIGH risk
- Most common pattern is NSIP, UIP
- Often occurs early in disease process (first 5-7 years)

- **Gastric antral vascular ectasia (GAVE)**

- Results in GI bleeding that can be life-threatening



Twitter @walinom

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Scleroderma Renal Crisis (SRC) vs TTP

	SRC	TTP
Clinical presentation	Dyspnea, altered mentation, hypertension (not always!) Clinical features of scleroderma	Fever, HTN, purpuric rash, bleeding, neurologic symptoms
Typical laboratory findings	MAHA, AKI, proteinuria, hematuria	MAHA, thrombocytopenia , AKI, proteinuria, hematuria
Other possible labs	Scleroderma autoantibodies	Low levels of ADAMTS13
Treatment	ACE inhibitor (captopril)	PLEX, steroids, rituximab

Raynaud's Phenomenon

Triphasic discoloration of the digits usually triggered by cold exposure
Pallor → *Cyanosis* → *Erythema*



Primary (common)

- F > M, onset age ~20's, often family history of Raynaud's
- Symmetric, typically not progressive
- Predictable onset (cold)
- **ANA negative**

Secondary = Due to underlying CTD/Vasculitis/Infection

- Asymmetric, can be progressive
- **Abnormal nail folds**
- Pits and ulceration of digits
- Males, onset >40's
- Thumb involvement

Secondary Causes of Raynaud's:
SSc, MCTD, SLE
HypoT4, Carcinoid, PCC
HBV, HCV, ParvoB19
Heme malignancy

On exam:

- Look for features of CTD
- Examine nail fold capillaries

Labs (only needed if suspicious of secondary):

- CBC, Cr, U/A, LFT, CK, ESR, CRP
- ANA, ENA, RF, C3/4, Cryos
- TSH, SPEP/UPEP
- HBV, HCV
- PTT, APLA

Treatment:

- Conservative measures
- **1st line - Calcium channel blockers**
- Can consider topical nitrates, PDE5 inhibitor

MCQ 4

A 45-year-old woman presents with a 10-year history of Raynaud's phenomenon, weight loss over the past 2 years and difficulty breathing. Physical exam shows skin thickening extending to the fingers and crackles on lung auscultation. Inverse PFTs show FVC 65% predicted, normal biventricular function with

Which of the following is the most likely diagnosis?

- A. Pulmonary arterial hypertension
- B. B. Interstitial lung disease**
- C. C. Left-sided heart failure
- D. D. Chronic aspiration pneumonia

This patient likely has diffuse systemic sclerosis which is more associated with rapid onset of Raynaud's and organ involvement along with positive anti-Scl70. PFTs showed reduced FVC **disproportionally reduced DLCO** which is typical of ILD> PAH would present with isolated DLCO and normal FVC. Left sided heart failure would present with reduced EF. Chronic aspiration is possible in SSc due to esophageal dysmotility but is less consistent with this pattern of PFT. Of note, PAH ECHO features: RA dilatation, RV dilatation, RV dysfunction, **flattened intraventricular septum**, pericardial effusion, elevated RVSP.

Myositis

Often acute, symmetric progressive weakness or proximal muscle groups

Myositis-specific Antibodies Syndromes

Anti-synthetase Syndrome = Anti-Jo1 Antibody

- Raynaud's phenomenon, mechanic's hands, arthritis

Anti-MDA5

- Rapidly progressive ILD, skin ulceration

Anti-NXP2 and Anti-TIF1- γ Antibodies

- Highly associated with malignancy

Ensure cancer screening UTD for **ALL DM/PM at presentation or any disease flare.*



Images: A, J Rheum; B, NEJM 2023.

Ddx: Myositis/Myopathy

- Drugs
 - Statins, colchicine, alcohol, etc
- Infectious/ Viral
 - HIV, influenza, EBV, CMV
 - Pyomyositis
- Hypothyroid myopathy
- Electrolyte disorders
 - Severe hypokalemia, hypophosphatemia
- Genetic myopathy
 - (eg muscular dystrophy, or disorders of glycogen/lipid metabolism, mitochondrial disorders)

Dermatomyositis/Polymyositis (DM/PM)

Clinical Features:

- **Muscle weakness:** Insidious over weeks/months, symmetric and **proximal > distal**, neck flexor
 - Can involve: Heart, diaphragm, oropharynx, and esophagus
- **Cardiac:** myocarditis, arrhythmias, CHF
- **Pulmonary:** ILD (NSIP, UIP) DLCO or CT abnormalities, Pulm HTN
- **Skin:** Gottron's papules, shawl sign, heliotrope rash, generalized erythroderma, periungal erythema, mechanic's hands, scalp psoriasiform changes, calcinosis cutis



Heliotrope Rash



Nailfold Capillary
Dilatations



Gottron's Papules

Dermatomyositis/Polymyositis (DM/PM)

Investigations:

- Labwork: **CK**, AST, LDH, inflammatory markers, **myositis specific antibodies**, ANA
 - Note: CK can be normal = amyopathic DM
- **Muscle MRI**: Demonstrates edema, can help guide biopsy

Muscle Biopsy = * Gold standard *

- Muscle EMG: Irritability, low amplitude, repetitive discharge
- Trop, ECG +/- Echo to rule out cardiac involvement
- SLP assessment – rule out oropharyngeal/esophageal involvement
- Spirometry with MIPs/MEPs – rule out diaphragmatic involvement
- Age-appropriate malignancy screening; consider CT chest/abdo/pelvis (solid tumour; adenocarcinoma)

DM can be associated with **malignancy** which typically presents **at time of diagnosis** of muscle weakness or **within 1 year** of diagnosis

DM/PM Management

1st line:

- Exercise!
- High po dose steroids (0.5- 1mg/kg)
 - Typically PO, however may use IV pulse if severe
- DMARD = steroid sparing agent
 - MTX, AZA, tacrolimus, MMF
 - Hydroxychloroquine helpful for skin manifestations only

*****Rapidly progressive ILD = needs triple therapy (IV steroids + 2 immunosuppressive options)***
-Ritux, CYC, IVIG, CNI, MMF

Refractory or Severe:

- IVIG
- Rituximab
- Cyclophosphamide
- Abatacept

Consider prophylaxis for steroid complications:

- Bisphosphonates*
- Calcium/Vit D*
- PPI, Septra*

Continue routine, age-appropriate malignancy screening as risk of malignancy elevated for at least five years following diagnosis of DM

Other Myosidities

Inclusion body myositis

- **Older, M > F**, insidious onset
- CK tends to be lower
- **Distal > proximal muscle weakness**
- Poor treatment response

Immune Mediated Necrotizing Myopathy (IMNM)

(aka Necrotizing Autoimmune Myositis (NAM))

- Severe myopathy
- CK markedly elevated
- Persistent after d/c statin
- Absence of skin manifestations
- ***Anti-HMG CoA reductase antibody*** [up to 50% are actually statin naïve]
- *Rule out a paraneoplastic syndrome*

A word on statins....

Statin muscle-related adverse events include:

- Myalgia: Normal CK, improves with d/c
- Myopathy: Weakness with OR without an elevation in CK, typically improves with discontinuation
- Myositis (including IMNM): Immune-mediated event, will require immunosuppression
- Myonecrosis
- Clinical rhabdomyolysis: Supportive Rx

Screening for ILD in Patients with Rheumatic Disease

ACR Guideline 2023

- HIGHEST risk SARDs: RA, Systemic sclerosis, Idiopathic inflammatory myositis (incl PM, DM, MDA5, IIM), MCTD, Sjogren's

– Who to screen? High risk for ILD, if no high risk features → then test in presence of ILD sx

Risk Factors	SS	RA	IIM	MCTD	Sjogren's
	Scl-70+, ANA with nucleolar pattern, M, AA, elevated APR, early disease (5-7Y)	Higher RF, CCP+, Smoker, older age of onset, M, Higher BMI Higher disease activity	Anti-synthetase, mechanic's, anti-Jo, MDA5, Pm/Scl, Ro52+, ulcerating lesions	Dysphagia RP SS lab or clinical features	Ro-52 RP Lymphopenia

Do NOT screen with CXR, 6MWT, lung biopsy, bronchoscopy!

Note: CTA is inadequate to diagnose ILD

- Screen with: PFT – need DLCO + TLC and/or CT (High Res CT + PFT >PFT alone)
 - If diagnosed – monitor with same tests used for diagnosis BUT add ambulatory desaturation (walking) test
 - No clear guidance on frequency of monitoring

Overview

1. Inflammatory Arthritis

- Rheumatoid arthritis
- Seronegative arthritis/Spondyloarthritis

2. Infectious

- Septic
- Gonococcal
- Lyme arthritis

3. Crystal Arthritis

- Gout
- Pseudogout

4. Connective Tissue Disease

- SLE
- Sjogren's Syndrome
- Systemic Sclerosis
- Myositis

5. Vasculitis

- Large vessel
- Medium vessel
- Small vessel

6. Inflammatory Disorders

- Still's Disease
- Relapsing Polychondritis

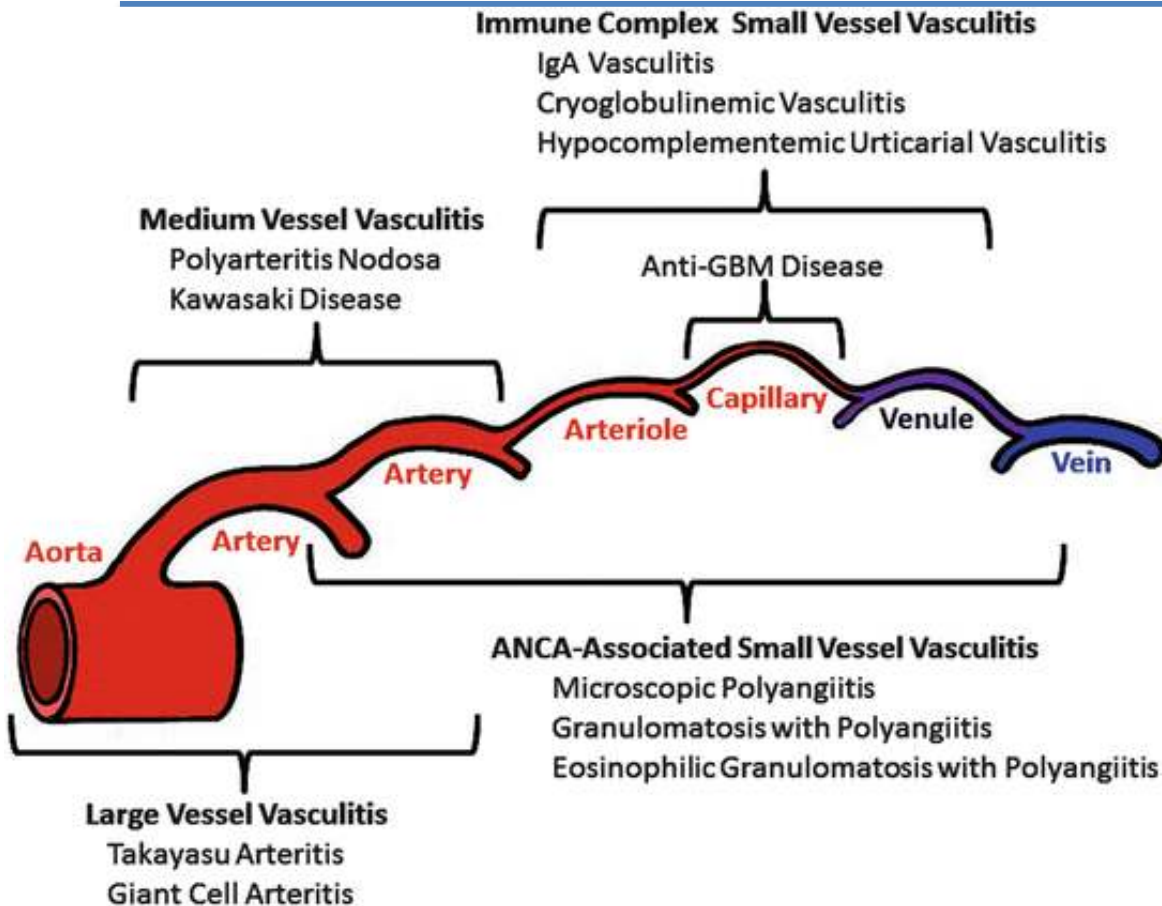
MCQ 5

A 29-year-old man presents with recurrent painful oral ulcers for the past 2 years. He has also developed several episodes of painful genital ulcers that healed with scarring. On examination, there are multiple aphthous ulcers in the oral mucosa and a healed scar on the scrotum. Eye exam reveals anterior uveitis. Blood work reveals normal CBC, renal and liver function. Serology negative for ANA, ANCA, ENA and APLA. A pathergy test revealed pustule formation at 72h.

Which of the following is the most appropriate initial treatment for this patient's disease?

- A. Colchicine
- B. Topical corticosteroids only
- C. Cyclophosphamide
- D. Anti-TNF therapy

2012 “Chapel Hill” Nomenclature of Vasculitides



Vasculitis: inflammation of any blood vessel → may lead to stenosis, occlusion, thrombosis, aneurysm or dissection

- ALL of which can lead to end organ damage and dysfunction from impaired blood flow

Secondary Causes of SVV

- Infection (Hep B, C, 3° syphilis, COVID)
- Drugs (levamisole – cocaine additive)
- CTD (SLE, RA, Sjogren's), sarcoid
- Malignancy

Variable Vessel Vasculitis

- Behcet's
- Cogan's (keratitis + sensorineural hearing loss)
- Thromboangiitis obliterans (Buerger's)

High Yield Vasculitis Overview

Vessel Size	Name	Characteristics
Large	Giant Cell Arteritis	New onset headache, jaw claudication, diplopia, amaurosis fugax
	Takayasu's Arteritis	Aortitis usually in Asian F, < 40y, "pulseless disease"
Medium	Polyarteritis Nodosa	HBV, mesenteric arteritis, neuropathy, B symptoms, orchitis
	Kawasaki's Disease	Childhood onset, conjunctivitis, cervical lymphadenopathy, strawberry tongue, rash (Warm "CREAM")
	Thromboangiitis obliterans (Buerger's)	Male smokers, necrotic digit tips, hand/foot claudication
Small	Granulomatosis with polyangiitis (GPA)	c-ANCA (PR3), pulmonary-renal, nasal and sinus granulomas
	Eosinophilic granulomatous polyangiitis EGPA	p-ANCA (MPO), eosinophilia, adult onset asthma, cardiac
	Microscopic Polyangiitis (MPA)	p-ANCA, pulmonary-renal, peripheral neuropathy
	IgA Vasculitis	IgA deposition--GI, skin, renal (may see a lot of proteinuria + purpura)
	Anti-GBM Disease	True pulmonary-renal syndrome, Rx PLEX until negative Ab level
	Cryoglobulinemic Vasculitis	Type 1 - MPNs, Type II/III—HCV(80%) or idiopathic
	Cutaneous Leukocytoclastic Vasculitis	Drug reaction, hypersensitivity

Large: Giant Cell Arteritis

LR's based on JAMA RCE

Symptoms:

Cranial Manifestations

- Headache (76% sensitive)
- Jaw claudication (**LR +4.2**)
- Diplopia (**LR +3.4**)
- Scalp tenderness
- Amaurosis Fugax/Vision loss
- Stroke (can involve posterior circulation ex. vertebral or posterior cerebral)

+/- Extra-cranial/Systemic Manifestations

- Limb claudication (upper or lower extremity)
- Constitutional symptoms



Eye Wiki: Dr N Volpe

Physical Examination:

- Beaded, prominent or tender temporal arteries (LR + 4.6, 4.3, 2.6),
- Absent temporal artery pulses
- Listen for bruits in all major vascular territories
- Assess bilateral BPs and pulses (10mmHg differential in systolic BP)
- Perform fundoscopy (look for acute ischemic optic neuritis, pale optic disc) and visual acuity
- Consult ophtho for dilated exam

ACR/EULAR 2022 Classification Criteria for GCA: score of ≥ 6 needed

Note: These are criteria which should be applied once you've made a diagnosis of vasculitis... alternate dx mimicking vasculitis should be excluded before applying criteria

Absolute Criteria: Age $\geq 50y$

Clinical	Morning stiffness in shoulders and neck	2
	Sudden vision loss	3
	Jaw/tongue claudication	2
	New temporal headache	2
	Scalp tenderness	2
	Abnormal temporal artery (decreased pulse, tender, cord-like/firm)	2
Lab or Imaging	ESR ≥ 50 or CRP $\geq 10\text{mg/L}$	3
	Positive temporal artery biopsy or halo sign on ultrasound	5
	Bilateral axillary involvement (stenosis, occlusion or aneurysm on CTA, MR, Angio or Ultrasound, halo sign on US)	2
	PET scan activity throughout aorta	2

Large: Giant Cell Arteritis

Investigations:

- ESR (rule out: Normal ESR **LR 0.2**)
- US of the temporal arteries & large vessels
 - Imaging of temporal arteries (halo sign, non-compressible artery)
 - *Not recommended over bx in ACR 2021, however EULAR 2022 suggests that US + high pre-test probability is sufficient for Dx*
- Temporal artery biopsy- ideally obtain unilateral bx within 14 days of starting steroids (Sn 87%)
- Once diagnosis is **confirmed**:
 - CTA or MRA of neck, chest, abdomen and pelvis to r/o large vessel involvement at baseline

2023 EULAR Imaging Recommendations:

US now recommended as 1st line in patients with GCA and include axillary arteries. FDG-PET or MR can also be used

For Takayasu's – MR is preferred imaging modality (PET/US as alternatives)

Imaging is not routinely recommended for FU – US, PET or MRA can be used for suspected relapse.

MRA/CTA/US for long term monitoring of structural damage.

Large Vessel Vasculitis

Management of GCA:

Visual symptoms/loss or critical cranial ischemia:

- IV pulse steroids 1g x 3 days then
- Prednisone 1mg/kg daily + Tocilizumab (TCZ)

No visual symptoms/loss or critical cranial ischemia:

- Prednisone 1mg/kg daily + Tocilizumab

Extracranial GCA:

Prednisone 1mg/kg daily + (TCZ or MTX)

ASA only if critical/flow-limiting lesion of carotid/vertebral arteries

Takayasu Tool Box:

- **Clinical Diagnosis:** HTN, SBP>20 between arms, angina/MI, constitutional, pulseless/bruits + imaging (MRA or CTA). No specific biomarkers. New Classification criteria 2022 (don't memorize)
- **MRI is preferred imaging modality**
 - PET/US as alternatives
- Imaging is not routinely recommended for FU – US, PET or MRA can be used for suspected relapse.
- Biopsy impractical on large artery!
- **Treatment:** GC + non GC immunosuppressant ex MTX, AZA, TNFi

Polymyalgia Rheumatica (PMR)

- **Diagnosis (2012 EULAR/ACR): *required criteria**

- Age > 50y*
- Elevated ESR or CRP*
- Bilateral shoulder ache* +/- hip pain stiffness, absence of other joint pain.
- Morning stiffness > 45 minutes duration
- Negative RF and CCP (and assumes normal CK → If elevated, think myositis!)
- Ultrasound criteria: Shoulder/hip bursitis/tenosynovitis/synovitis

Relationship between GCA and PMR:

- ~50% of patients with GCA will have PMR
- ~15% of patients with PMR will have GCA

- **Treatment (2015 EULAR-ACR guidelines):**

- Prednisone 12.5-20mg/d x 2-4 weeks
- Then taper to 10mg/d within 1-2 months if response
- If relapse, increase to pre-relapse dose, then decrease gradually and/or add DMARD (e.g. methotrexate)

***NEW FDA Approval for Sarilumab** (IL-6 receptor inhibitor) approved for use in PMR patients who cannot tolerate GC taper
-interferes with the ability to produce an ARP (CRP, ESR) thus reducing the impact of these biomarkers on clinical assessment of disease activity

Medium: Polyarteritis Nodosa (PAN)

Necrotizing inflammation of medium sized or small arteries

WITHOUT glomerulonephritis → can have renal infarcts

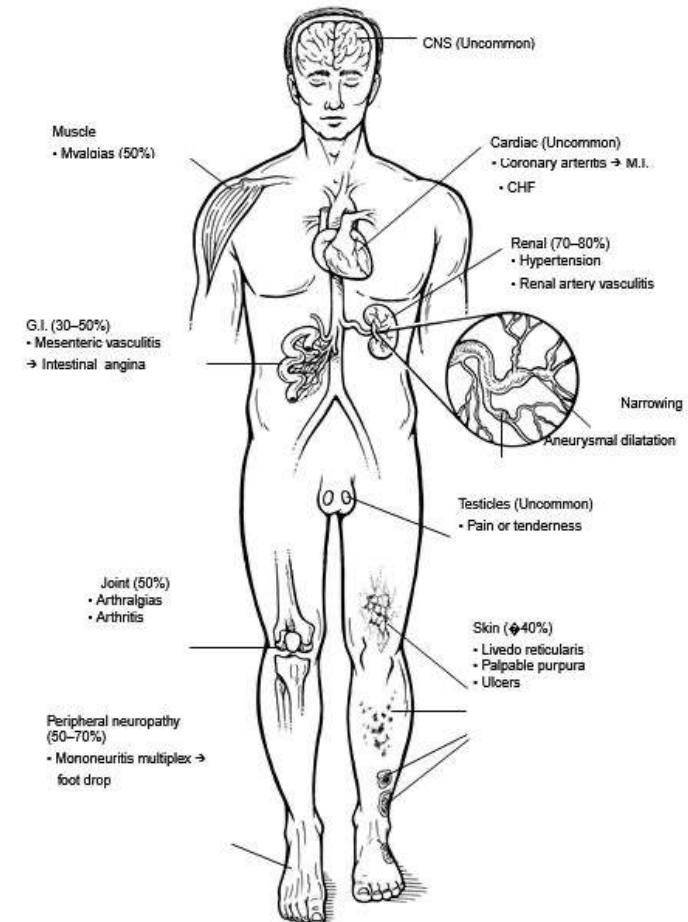
Clinical Presentation:

- Constitutional Symptoms: Fever, weight loss >4kg
- Cutaneous: Nodules, Ulcers, Livedo reticularis
- Testicular pain or tenderness
- Abdominal pain (post-prandial)
- Myalgias (excluding shoulder/hip girdle)
- Inflammatory arthritis
- Mono/polyneuropathy or mononeuritis multiplex
- Arteriographic abnormalities: Aneurysms or stenotic lesions in mesenteric/hepatic/renal arteries + branches
- Used to be predominantly hep B mediated

Treatment:

- If HBV-related: GC, anti-virals +/- PLEX
- Idiopathic: GC + MTX/AZA OR in severe disease high dose GC + CYC

Classically, PAN is serologically negative (ANCA, RF), but can have positive serology. Go by clinical presentation to make the diagnosis!



ANCA Associated Small Vessel Vasculitis

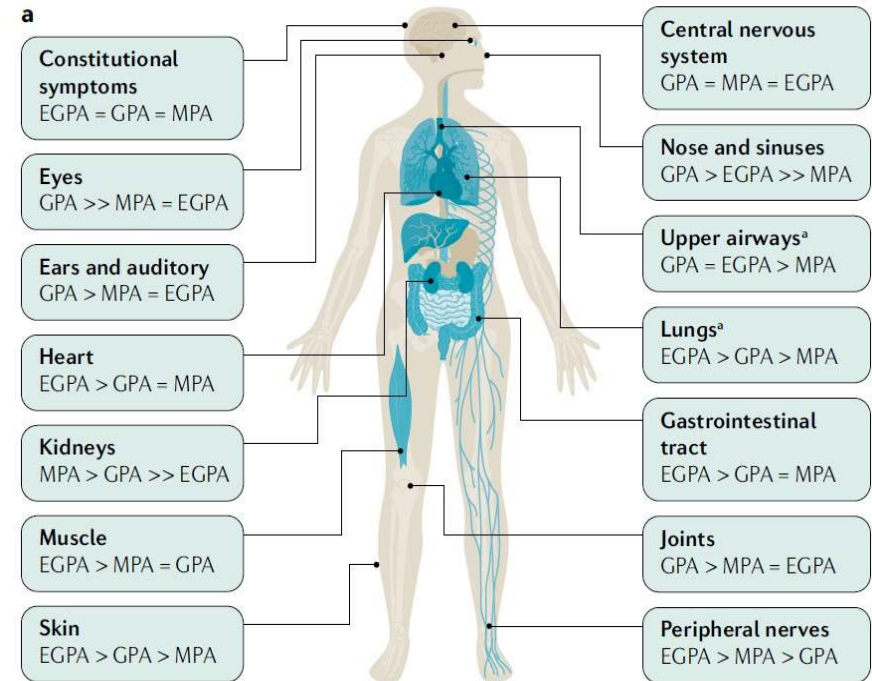
	GPA	MPA	EGPA
Serology	Usually c-ANCA (PR3) positive (80%)	More p-ANCA positive (65%)	Can be ANCA +/- If ANCA+, usually – p-ANCA (40%)
Overview	Pulmonary renal syndrome with ENT involvement - Epistaxis, nasal ulcers, saddle nose deformity - Subglottic stenosis, cough	Pulmonary-renal without ENT Lung involvement is different too Capilliritis, alveolar bleeding May have renal limited MPA	Least common Adult onset asthma, allergic rhinitis, peripheral eosinophilia, neuropathy



Saddle nose deformity



Granuloma



Source: ACR Image Gallery

Management of GPA/MPA

Induction Therapy (severe disease*):

Glucocorticoids: 500-1000mg IV or 1mg/kg PO x 3-5 days then **reduced dose taper**
AND

Rituximab: 375 mg/m² IV weekly x 4 weeks OR 1000mg IV Day 1, Day 15

Cyclophosphamide (CYC) can be used as an alternative if contraindication or failed rituximab → *in contrast to KDIGO that prefer CYC for induction if RPGN with Cr >354*

Maintenance: continue Ritux or alternative DMARD (**azathioprine – KDIGO**)

- Generally AAV is monophasic → usually continued treatment for >18mo
- If relapsed disease use alternative agent

Induction Therapy (non-severe disease): Glucocorticoids + MTX/AZA/MMF

***Organ/Life Threatening Disease in AAV**

- GN
- Pulmonary hemorrhage
- CNS involvement
- retro-orbital disease
- cardiac involvement
- mononeuritis multiplex
- mesenteric involvement

Avacopan – novel agent, C5a receptor inhibitor for steroid sparing effect **only for GPA/MPA**

ADVOCATE Trial: improved renal recovery in pts with GN with lower GC exposure/AE's. (NEJM, 2023 384:599-609)
EULAR: In combo with Ritux/CYC may be considered for induction to reduce steroid exposure

Management of GPA/MPA – to PLEX or not to PLEX?

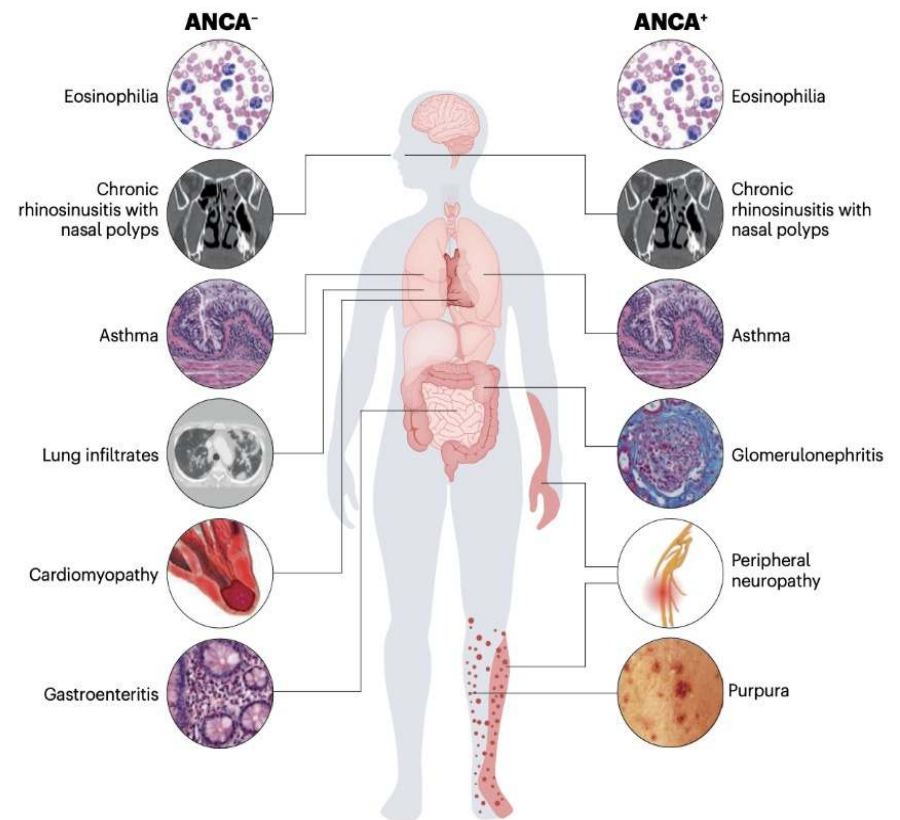
“Urgent plasma exchange is NOT recommended as part of initial induction therapy for most adult patients with severe GPA or MPA – CANVASC Guidelines, 2020

- ❖ **Recommended against in pulmonary hemorrhage** unless as salvage therapy in critically unwell (eg after failure of GC + Ritux or Cyclophos) – ACR/EULAR
- ❖ **Recommended *against* in RPGN** unless high risk of ESRD and accept risk of infection → *compare to KDIGO 2024: “consider” PLEX if Cr >300, patients requiring dialysis or with rapidly rising Cr, or patients with diffuse alveolar hemorrhage with hypoxemia due to active GN*
- ❖ **PLEX recommended for concurrent anti-GBM disease – ACR/EULAR/KDIGO/CANVASC all agree**

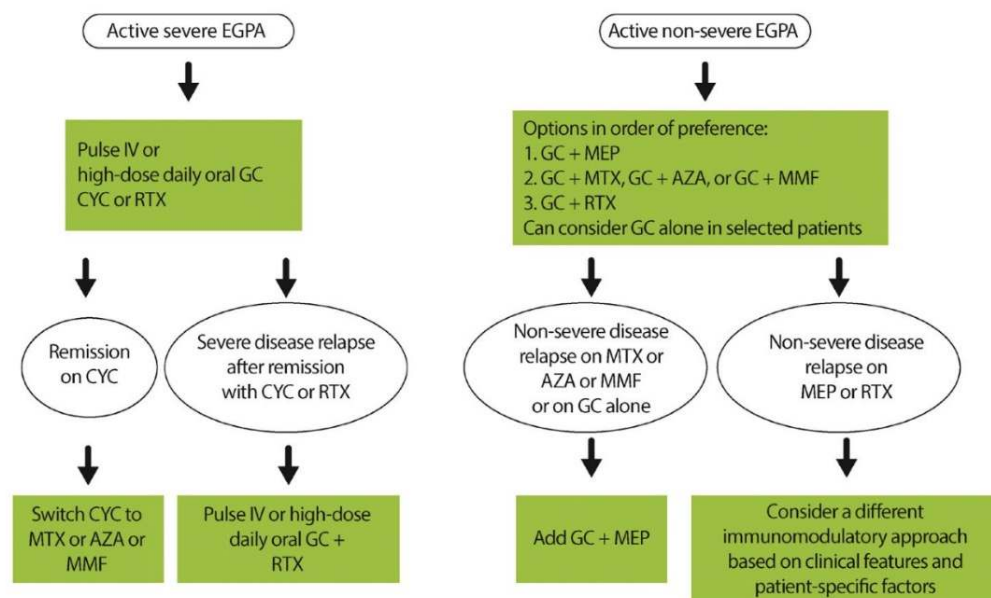
ANCA vasculitis with severe kidney disease	Vasculitis with diffuse pulmonary hemorrhage	Vasculitis in association with anti-GBM antibodies
Seven treatments over a maximum of 14 days, 60 ml/kg volume replacement, albumin substitution	Daily until bleeding stops, replace albumin with fresh, frozen plasma	Daily for 14 days or until anti-GBM antibodies are undetectable

Eosinophilic Granulomatosis with Polyangiitis (EGPA)

- Least common of the AAV
- Suspect in a patient with history of adult-onset asthma presenting with peripheral eosinophilia and vasculitis sx
- Disease presentation can be different depending on ANCA +/-



Management of eGPA



Five-Factor Score should be used to guide therapy

- Proteinuria >1g/day
- Creatinine > 138.7 umol/L
- GI tract involvement
- Cardiomyopathy
- CNS involvement

Score ≥ 1 = severe disease

Revised criteria → used for prognosis, continue to use OG FFS for treatment decisions

- Age 65+
 - Cr >150
 - GI involvement
 - Cardiac involvement
 - Lack of ENT involvement
- Worse prognosis in ANCA - patients

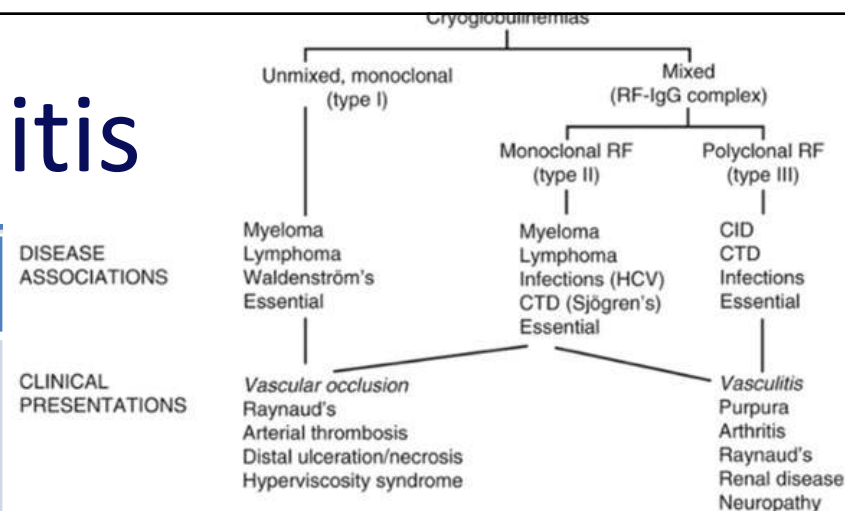
Disease Relapse

- If severe disease, previously induced with Cyclophosphamide: **Rituximab**
- If severe disease, previously induced with Rituximab: **Cyclophosphamide**
- If non-severe relapsing disease while on MTX/AZA/MMF/GC monotherapy: Add mepolizumab

*Note due to lack of RCTs in this population most of this guidance is “ungraded position statement” (consensus)
2021 ACR Guidelines for the Management of Small Vessel Vasculitis. *Arthritis Care Res.* 73(8):1088-1105

Cryoglobulinemic Vasculitis

Type I	Type II/III
Seen in clonal hematologic disease (MGUS, MM, CLL, WM)	Can be secondary to chronic infection (Hep C in 50% , Hep B, HIV, IE), SARD (Sjogren's), or lympho-proliferative disorders
Clinical manifestations due to direct vascular occlusion in small/medium vessels	Clinical manifestations due immune complex deposition in small/medium vessels leading to vasculitis
Refer to heme to treat underlying heme disorder +/- PLEX	HCV Negative: Treat underlying disease Non-Severe: GC +/- colchicine Severe: GC +/- Ritux Life-Threatening: GC + Ritux + PLEX HCV Positive: ALL patients get anti-viral therapy! Moderate/Severe Disease: may add GC +/- Ritux Life-Threatening: GC + Ritux + PLEX



CTD = connective tissue disease; CID = chronic inflammatory disease.

Mild/Moderate: Non-ulcerating skin lesions (purpura, acrocyanosis, livedo reticularis), non-debilitating peripheral neuropathy, arthralgias/arthritis

Severe: Cutaneous ulcers, progressive or debilitating neuropathy, GN with renal failure or nephrotic syndrome

Life Threatening: Rapidly progressive GN, CNS involvement, intestinal ischemia, DAH



Cryoglobulinemia
Source: Dermnetnz.org

Behcet's Disease

Image source: ACR Image Bank

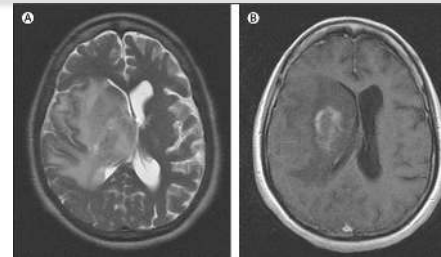
- Variable vessel vasculitis

TABLE 1. International Criteria for Behçet's Disease

Symptom	Score*
Oral ulcer	2
Genital ulcer	2
Ocular manifestation	2
Skin manifestation	1
Vascular manifestation	1
Central nervous system involvement	1
Positive pathergy test	1

Scoring at least 4 indicates Behçet's disease diagnosis. Pathergy testing is optional, but when it is performed, it is given a score of 1.

- HLA-B51 seen in 40-60% of patients, more severe disease
 - More prevalent in those from Mediterranean basin, where there is higher proportion of HLA-B51 thus not as helpful in diagnosis



- CNS manifestations: large inflammatory lesions, pyramidal signs, dural sinus thrombosis
- Vascular involvement: VTE, pulmonary or peripheral aneurysms

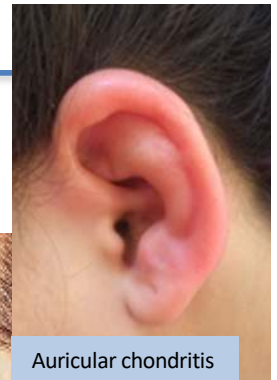


Treatment: based off of system involvement

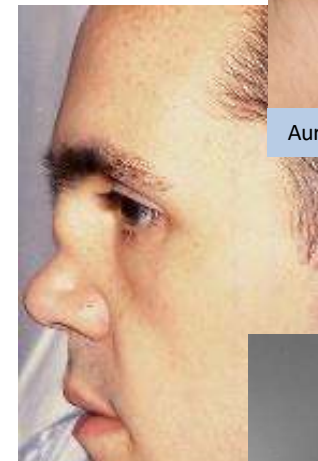
- Ulcers: topical GC, colchicine, apremilast, AZA, TNFi
- Aneurysms: surgery, CYC, Infliximab
- CNS involvement: AZA, TNFi
- Controversy over VTE management! AC appears to confer NO additional benefit, immunosuppression preferred tx

Relapsing Polychondritis

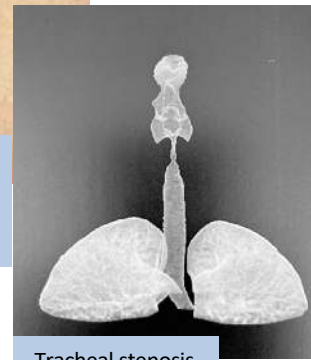
- Rare autoimmune disorder
- Recurrent episodes of inflammation of the cartilage typically presenting as chondritis of the nasal bridge, ear, or hyaline cartilage of the bronchial tree
- Can also have ocular manifestations (episcleritis, scleritis)
- MSK manifestations are common with asymmetric, seronegative arthritis (often involves SCM, costochondral or manubriosternal joints)
- NO specific diagnostic test → often biopsy of affected tissue is required
- Tx: Based on organ involvement, may necessitate surgery (i.e. tracheal stenosis, aneurysm repair)
 - NSAIDs, GC, MTC, AZA, CYC



Auricular chondritis



Nasal bridge involvement with resultant saddle nose deformity



Tracheal stenosis

MCQ 5

A 29-year-old man presents with recurrent painful oral ulcers for the past 2 years.

He has a history of recurrent genital ulcers, skin lesions (papulopustular rashes, EN like rashes, PG...), CNS involvement, vascular manifestations and pathergy. Treatment is based on disease manifestations. Colchicine is first line treatment for ulcers. He has history of uveitis but none currently thus does not require topical GC or anti-TNF. Ocular involvement may warrant immunosuppression, especially if posterior segment involvement and treatment can be guided in conjunction with ophthalmology. CYC or TNF would be reserved for more severe disease involvement.

Which of the following is the best treatment for this patient's disease?

- A. Colchicine
- B. Topical corticosteroids only
- C. Cyclophosphamide
- D. Anti-TNF therapy

Overview

1. Inflammatory Arthritis

- Rheumatoid arthritis
- Seronegative arthritis/Spondyloarthritis

2. Infectious

- Septic
- Gonococcal
- Lyme arthritis

3. Crystal Arthritis

- Gout
- Pseudogout

4. Connective Tissue Disease

- SLE
- Sjogren's Syndrome
- Systemic Sclerosis
- Myositis

5. Vasculitis

- Large vessel
- Medium vessel
- Small vessel

6. Auto-Inflammatory Disorders

- Still's Disease/MAS
- Relapsing Polychondritis

Autoinflammatory disorders

- Disorders of **innate immunity** → recurrent inflammation **without high-titer autoantibodies or antigen-specific T cells** (i.e., *not autoimmune or adaptive immunity*)
- Driven by **cytokine dysregulation** (IL-1, IL-6, IL-18, TNF- α)
- Characterized by fever, elevated inflammatory markers
- Ferritin in the 1000s? → think about possible autoinflammatory disease
- Examples of autoinflammatory disorders you may have heard of:
 - Still's disease*
 - Macrophage activation syndrome*
 - VEXAS syndrome
 - Vacuoles, E1 enzyme, X linked, Autoinflammatory, Somatic mutation
 - First described in 2020! See bonus slides for info
 - Cryopyrin associated Periodic syndromes (CAPS)
 - Familial Mediterranean fever (FMF)

*higher yield to know of for exam

Still's Disease

Still's Disease =
formerly sJIA and
AOSD

- Nonfamilial systemic autoinflammatory disorder
 - More common in younger adult, F>M
- Fever – daily, quotidian*
- Arthralgias/arthritis – usually asymmetric, oligo
- Salmon pink evanescent rash
- Elevated ferritin, leukocytosis, inflammatory markers
- Other features: sore throat, lymphadenopathy, hepatosplenomegaly, abnormal LFTs

Still's Disease Diagnosis

Fever >39C for at least 7 days

Rash – transient, coincides with fever spikes, usually truncal

MSK involvement – myalgia, arthralgia

HIGH levels of inflammation – neutrophilic leukocytosis, elevated CRP, ferritin

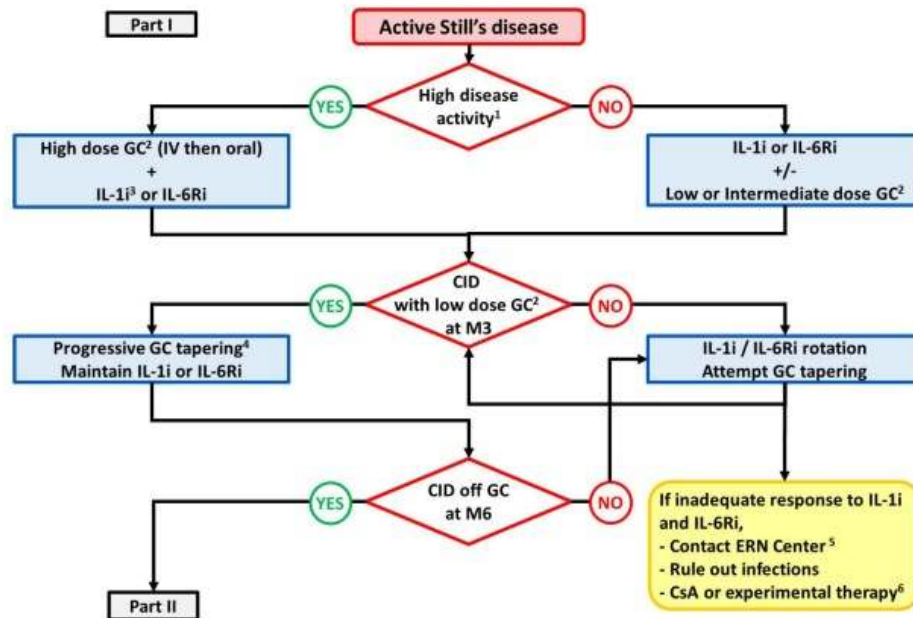


Source: Dr McKenna Office

*DDX: quotidian fever

- Still's disease
- Kawasaki
- Malaria
- Meningococcal IE
- Miliary TB

Still's Disease



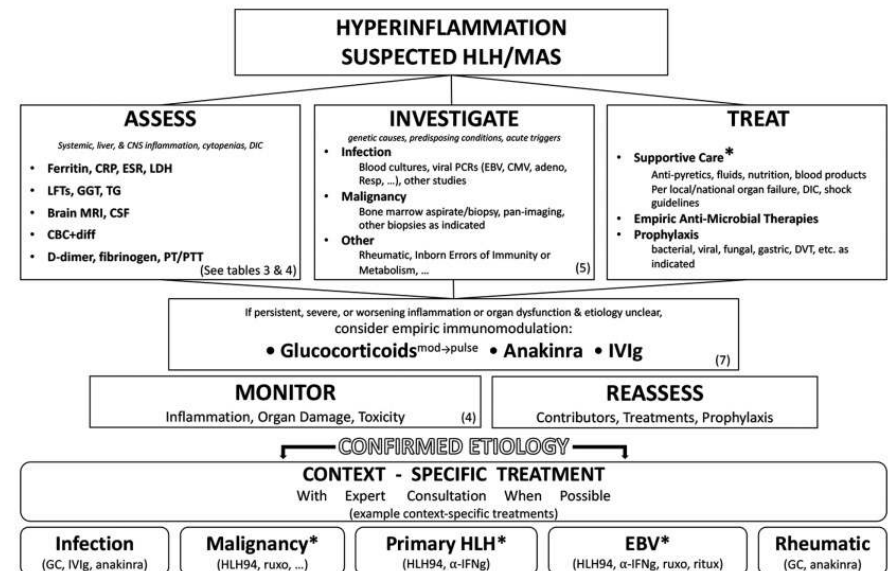
- Early initiation of IL-1 or IL-6 inhibitor is recommended
- Avoid prolonged GC use
- CID = clinically inactive disease
 - Always ensure disease is stable off GC prior to tapering any biologic

May still see use of NSAIDs for mild disease (non-disabling fever, rash, arthralgias or mild arthritis, no MAS), but this is not a recommended strategy. csDMARDs (i.e. MTX) no better than placebo. **bdMARDs targeting IL- 1 or IL- 6 are the treatment of choice.** csDMARDs should be considered if unable to access IL- 1/6i.

Macrophage Activation Syndrome (MAS)

LIFE threatening autoinflammatory syndrome caused by excessive immune system activation

- MAS is secondary hemophagocytic lymphohistiocytosis (HLH) occurring in context of rheumatic disease (Still's disease, SLE...)
- Characterized by persistent fever, splenomegaly, elevated ferritin (>10,000), cytopenias, abnormal LFTs, intravascular coagulation and elevated/rising triglycerides
- Tx: high dose steroids + IL-1, CNI, IFN- γ



Rheumatologic Emergencies

- **Macrophage Activation Syndrome** → high dose steroids + IL-1, CNI, IFN-inhibitor
- **Septic joint** → Antibiotics, Ortho for washout
 - If can't go to OR and large joint, daily aspirations with synovial WBC
- **GCA with vision loss** → Pulse steroids + Tocilizumab
- **Pulmonary-renal syndrome**
 - **If ANCA+** → Steroids + Rituximab (or Cyclophosphamide)
 - **If anti-GBM+** → Definite PLEX until negative Ab levels, Steroids + CYC/Ritux
- **Scleroderma renal crisis** → ACE inhibitors (Captopril)
- **Thrombotic microangiopathies** (antiphospholipid syndrome, SLE) → Anticoagulation (caution if hemorrhage component), tailor treatment to clinical presentation and cause
- **Systemic disease flare** (RA, SLE, polymyositis, seronegative, vasculitis)
 - Disease: Cardiac, pulmonary, renal, muscle, CNS/neurologic
 - Treatment complications: cytopenias, pneumonitis, renal/hepatic dysfunction, allergy etc

Extra Resources

American College of Rheumatology Clinical Practice Guidelines

- www.rheumatology.org/Practice-Quality/Clinical-Support/Clinical-Practice-Guidelines

Canadian Rheumatology Association- Position papers and practice guidelines

- <https://rheum.ca/resources/publications/>

Rheuminfo.com

- Canadian site by Dr. Andy Thompson, rheumatologist at Western University. Helpful patient handouts on diseases, drugs – good overview if practicing for RCE counselling station (e.g. “Counsel this patient on steroids / methotrexate ...”)

Physical Exam Videos and other learning resources

- **RheumTutor.com** - developed by Dr. Raj Carmona, a rheumatologist at McMaster
- **Podcast: Rheum for the Royal College** – developed by Dr Karim Ladak (more high level, use for topics if there is confusion)

ACR Image collection:

- <http://images.rheumatology.org/>



App: RheumaHelper – good for looking at classification criteria for each disease

Helpful Social Media Accounts for learners:

@Janetbirdope – Dr Janet Pope, Rheum @ U of T – helpful rheum #clinicalpearl posted weekly (physical exam findings!)

@BrownHospMed – Brown University twitter account – great cases with images

TikTok: @Rheum101

Questions?

- **RHEUM BONUS SLIDES FOLLOW:**
 - Summary of Autoantibodies in Rheumatology
 - More on immunosuppressive drugs
 - Approach to the hand exam/physical exam findings
 - Cardiovascular risk in rheumatology patients
 - Osteoarthritis
 - Fibromyalgia
 - IgG-4 Related Disease
 - VEXAS
 - Inclusion Body Myositis
 - Choosing Wisely Rheum Edition
 - Additional MCQs
- **ALLERGY/IMMUNOLOGY SLIDES are at END of Sunday course book!**
 - There will likely be 1+ MCQ on your exam

Autoantibody Summary

Autoantibody	Typical Disease Associations	Autoantibody	Typical Disease Associations
Anti-nuclear Antibody (ANA)	SLE Mixed Connective Tissue Disease Rheumatoid Arthritis	Anti-neutrophil cytoplasmic antibody (ANCA)- Myeloperoxidase (MPO/p-ANCA)	Vasculitis- Eosinophilic Granulomatous Polyangiitis (eGPA) Vasculitis- Microscopic Polyangiitis (MPA)
dsDNA	SLE	Anti-neutrophil cytoplasmic antibody (ANCA)- Proteinase 3 (PR3/c-ANCA)	Vasculitis- Granulomatous polyangiitis (GPA)
Anti-Smith	SLE	Rheumatoid Factor (RF)	Rheumatoid Arthritis Sjogren's, SLE
Anti-Ro/SSA	Sjogren's SLE	Anti-Cyclic Citrullinated Peptide Antibody (CCP)	Rheumatoid Arthritis
Anti-La/SSB	Sjogren's SLE	Anti-Glomerular Basement Membrane (GBM)	Anti-GBM Disease (Goodpasture Syndrome)
Anti-centromere	Limited Cutaneous Systemic Sclerosis (CREST)	Anti-Jo-1	Polymyositis/Dermatomyositis
Scl-70 (Anti-topoisomerase-I)	Diffuse Systemic Sclerosis	Anti-Mi2	Dermatomyositis
Anti-U1-RNP	Mixed connective tissue disease SLE	Anti-NXP2 Anti-TIF1-γ	Malignancy associated Dermatomyositis
Anti-histone	Drug induced SLE		

More on Drugs

Drug	Side Effects	Monitoring
Cyclophosphamide	Infertility (men and women), myelosuppression, infection (PCP), hemorrhagic cystitis, malignancy (bladder, skin, MPD), alopecia	Bi-weekly CBC
TNF- α inhibitors e.g. adalimumab	Infections (TB, increased frequency of typical infections), Malignancy (non-melanoma skin, ?lymphoma), demyelinating diseases, CHF (worsens if class III/IV), drug-induced lupus, infusion reactions	Prior to therapy: TBST, CXR HBV, HCV Live vaccines to be administered PRIOR to Rx
Colchicine	Diarrhea, nausea, myelo-suppression, myopathy, peripheral neuropathy (more common in CKD), drug interactions (CYP-3A4 inhibitors etc.)	None
Allopurinol	Gout flare, skin rash, nausea, diarrhea, transaminitis, allopurinol hypersensitivity syndrome (HLAB5801+), cytopenias, drug interactions (AZA, 6-MP)	Unclear

More on More Drugs

Drug	Side Effects	Monitoring
JAK inhibitors e.g. tofacitinib	Infections (TB, increased frequency of typical infections), Malignancy (non-melanoma skin, ?lymphoma), Herpes zoster, DVT/PE?, Lower GI perforation Sept 2021 FDA issued black box warning against all JAK inhibitors due to post-marketing surveillance safety trial showing increased risk of all cause mortality, MACE, thrombosis, and cancer	Prior to therapy: TBST, CXR, HBV, HCV Herpes zoster (shingles) vaccine to be administered PRIOR to Rx Live vaccines to be administered PRIOR to Rx
IL-12/23 antagonists e.g. Ustekinumab	Infections (TB, increased frequency of typical infections), Malignancy, Posterior reversible encephalopathy syndrome, Upper Respiratory infection	Prior to therapy: TBST, CXR HBV, HCV Live vaccines to be administered PRIOR to Rx
IL-17A inhibitors e.g. Secukinumab	Upper respiratory tract infections, Infections (Oral herpes), Inflammatory bowel disease?	Prior to therapy: TBST, CXR HBV, HCV Live vaccines to be administered PRIOR to Rx

The Hand Exam*

	OA	RA	Psoriasis	CPPD	Gout	SLE
Deformity	Bouchard's (PIP nodes) Heberden's (DIP nodes) Squaring of CMC (thumb)	Swan neck, Boutonniere, Z-deformity at thumb MCP subluxation MCP ulnar deviation Prominent ulnar styloid	Dactylitis	--	Tophi	Swan neck – if diffuse think Jacoud's arthropathy
Distribution	CMC PIPs, DIPs	Wrists MCPs, PIPs (NOT DIP)	DIP , PIP > MCP	MCP, radiocarpal (also elbows, shoulders)	ANY	MCPs
Skin	--	Vasculitis (rare) Rheumatoid nodules	Psoriasis	--	Tophi – check ears, elbows	Malar rash, discoid, alopecia
Nails	--	Rare - Nail ridging, clubbing	Pitting, leukonychia, onycholysis, splinter hemorrhages, oil drop	--	--	--
X-RAY Changes	Eccentric joint narrowing Osteophytes Subchondral sclerosis and cysts	Concentric joint space narrowing, Osteopenia Erosions	Marginal erosions, pencil and cup deformity Periostitis	Chondrocalcinosis involving TFCC	Erosions with overhanging osteophytes Tophi	Juxta-articular osteopenia Non-erosive

Non-Inflammatory Arthritis

OA: most common degenerative joint disease

- RF: F, age, obesity, prior trauma, smoking, repetitive load

Clinical Features: gradual onset, dull pain worse with activity, lack of EAM

- Note: can have inflammatory OA – central erosions, swelling (still no benefit from DMARDs)
- Mimicker of inflammatory arthritis

Non-Pharm Management for Hip/Knee/Hand OA

Strongly Recommended:

- Exercise +/- weight loss to reduce joint load
- Self-efficacy/management programs
- Gait aid, Tai Chi
- Knee brace, 1st CMC orthosis

Conditionally Recommended:

- CBT, therapy, thermal interventions, acupuncture, balance training, K tape, paraffin wax

Pharmacologic Management

Strongly Recommended:

- Oral NSAIDs, topical NSAIDs, IA steroids

Conditionally Recommended:

- Tylenol, Tramadol*, Duloxetine, Chondroitin

Strongly recommended AGAINST:

- TENS, bisphosphonates, glucosamine, DMARDs, non-tramadol opioids, PRP, stem cell/hyaluronic injections, arthroscopy

Consider surgery (TJA) if refractory sx to medical tx

New recommendation: do NOT delay TJA to pursue non-operative tx if severe sx/deformity/bone loss or neuropathic joint, do NOT delay for weight loss but still strongly encouraged

Delay TJA – to improve glycemic control, achieve nicotine cessation or reduction

Fibromyalgia/Chronic Non-Inflammatory Pain

Compositive of pain and other bothersome symptoms (sleep disturbance, fatigue, cognitive dysfunction...)

Diagnostic Criteria:

- 1) Widespread pain index and symptom severity (SS) scale
- 2) Symptoms have been present at a similar level for at least 3 months
- 3) The patient does not have a disorder that otherwise explain the pain (do not over-medicalize the patient)

Symptom based management with pharm & non-pharm strategies with aim to ↓ symptoms and ↑ function

- Self-management strategies are essential – active participant in care, patient education
 - Non-Pharmacologic: exercise, CBT, sleep hygiene, Tai Chi
 - Exercise has best available evidence – any type
 - Pharmacologic: no perfect drug, use lowest dose + gradually increase, expect modest response
 - Consider combination of drugs, constantly re-evaluate risk vs benefit
 - WHO-step up ladder – tramadol if moderate/severe pain, strong opioids discouraged
 - NSAIDs – low dose, short use, cannabinoids for sleep
 - Antidepressants – TCA, SSRI, SNRI
 - Anticonvulsants – low dose
- **Tender points are not in diagnostic criteria anymore!**
- Patients often report associated fatigue, cognitive dysfunction, and non-restorative sleep
- Screening lab tests: CBC, ESR, CRP, CK, and TSH MUST be normal

Good for oral
scenario!

BONUS: Intra-Articular Injections*

- Most robust evidence in support of intra-articular corticosteroids
 - Most benefit derived if there is inflammatory component
- Indications:
 - Knee osteoarthritis, especially if accompanied by an effusion
 - Crystal arthritis (gout, CPPD)
 - Inflammatory arthritis (ie. RA)

***Note:** Should not generally be used for hand OA or hip OA (hip IA injections should be US guided)

- Risks: infection (1 in 10,000), injury to tendon/periarticular structures (tendon rupture), elevation in BS/BP, skin hypopigmentation, post injection pain flare
- Contra-indications:
 - Absolute:
 - Possible septic arthritis
 - Prosthetic joint
 - Overlying cellulitis or suspected bacteremia
 - Allergy to corticosteroid
 - Relative:
 - Brittle diabetes
 - Clotting/bleeding disorders or anti-thrombotic medications
 - Failure to respond to previous intra-articular therapies

Beware: Rapidly Destructive Hip Disease
Journal of Bone and Joint Surgery, 2021
(103(22): 2070-2079)

Hip corticosteroid injections were associated with rapidly destructive hip disease. Risk is highest following high dose (≥ 80 mg) and multiple injections."
RDHD = rapidly progressed joint space narrowing, osteolysis leading to collapse of femoral head

Cardiovascular Risk Reduction in Rheumatic Diseases

Patients with inflammatory rheumatic diseases have increased CV risk compared to general population

- CV risk assessment and management should be performed starting within 6 months of diagnosis and at regular intervals, at least q5 years (or more depending on risk) and following changes in anti-rheumatic therapy
- Glucocorticoid/NSAID use should be minimized as disease state permits
- Disease activity should be optimally controlled to reduce CV risk
- Diet, exercise, smoking cessation universally recommended.

Special considerations for individual disease states:

- RA
 - Total cholesterol and HDL-c should be used in risk factor prediction (TG and LDL-c tends to be lower)
- Gout
 - Serum uric acid <0.36 mmol/L can potentially lower risk of CV events and CV mortality (*No preference between allopurinol and febuxostat)
 - Avoid diuretics, favour CCB or losartan (uricosuric)
- SLE
 - **BP target of <130/80** should be *considered* in all pts with SLE
 - **ACEi or ARB*** should be prescribed for patients with lupus nephritis, if urine PCR >500mg/g or hypertensive
 - Treatment with HCQ may reduce the risk of CV events
- APS
 - Low dose ASA recommended for A) asymptomatic aPL carriers with high risk profile, B) SLE patient with high risk aPL profile, but no APS, and considered for SLE patient with low-risk aPL profile

***Hurst et al (Am J Med 2023 Mar; 136(3) 284-293)**
“RAS-modifying drugs reduced the risk of CVD in patients with SLE”

IgG4 Related Disease

Immune-mediated condition associated with fibro-inflammatory lesions that can involve any organ

- Predilection for salivary glands, pancreas, lungs, kidneys, aorta, retroperitoneum
- Elevated serum IgG4
- Gold standard: histopathology
 - Often relies on biopsy → lymphoplasmacytic infiltrate/storiform fibrosis with IgG4 cell positive staining
- Exclusion criteria: fever, no response to steroids, positive serology, peripheral eosinophils, splenomegaly, non-consistent pathology

Treatment: glucocorticoids

2019 ACR Guideline for IgG4 Related Disease Classification Criteria.

Inclusion Body Myositis

- More common in M (3:1)
 - Most common myopathy in 60+
 - Insidious onset proximal and distal weakness
 - Finger flexor weakness is hallmark
 - May also have asx CK elevation, isolated dysphagia
 - May have associated dysphagia, atrophy, facial weakness
 - Very marked morbidity
 - Muscle Bx: endomysial inflammation, rimmed vacuoles, protein aggregation
- Degenerative muscle condition aka **NO** role for immunosuppressive medications

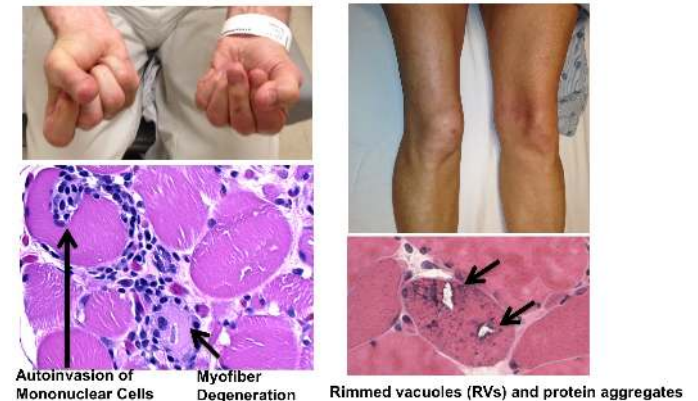


Image from John Hopkins

VEXAS

- X linked disease with somatic mutation in the UBA1 gene in blood cells and appears late in life (above 50yo)
 - Vacuoles – seen on BM of affected patients
 - E1 enzyme
 - X linked
 - Autoinflammatory
 - Somatic – acquired mutation
- Present with fever, systemic inflammation and organ inflammation
 - Inflammation may manifest as: skin – LCV, cartilage (auricular or nasal chondritis), lungs (alveolitis, serositis), blood vessel
- Often associated with macrocytosis (MCV >100) and myelosuppression – anemia, thrombocytopenia, lymphopenia with common association with MDS, MM, MGUS
- Increased risk of venous thromboembolisms

Choosing Wisely Rheumatology Edition

- Do NOT order ANA as a screening test in absence of specific signs/sx of SARD
- Do NOT order HLAB27 unless SpA is suspected clinically
- Do NOT order DEXA more often than every 2-3 years
- Do NOT prescribe OP treatment for low fracture risk
- Do NOT prescribe opioids for chronic rheumatic disease before optimizing non-opioid strategies
- Do NOT delay palliative symptom treatment in a patient with life limiting rheumatic disease
- Do NOT order labs for drug toxicity more often than every 8-12 weeks
- Do NOT order RF or anti-CCP unless clinically suspicious arthritis
- Do NOT perform serologic, imaging or genetic tests without checking for past results
- Do NOT dispose of regular waste in biohazardous waste containers when doing joint injection/aspiration

MCQ Bonus #1 2026

A 27 year old female has been admitted to the medicine ward with a 10 day history of fever, myalgias and new urticarial rash. She was started on broad-spectrum antibiotics, yet her fever has persisted. Initial blood cultures were positive for CONS. Blood work from today: Hb 85, Plts 60, WBC 2, ALT 84, AST 64, fibrinogen 0.8g/L, CRP 240, ESR 10, ferritin 18,000.

Which of the following is the most appropriate step in management?

- A. IV amphotericin
- B. IV high dose methylprednisolone
- C. PLEX
- D. Rituximab
- E. Cyclophosphamide

MCQ Bonus #1₂₀₂₆

A 27 year old female has been admitted to the medicine ward with a 10 day history of fever, myalgias and new urticarial rash. She was started on broad-spectrum antibiotics, yet her fever has persisted. Initial blood cultures were positive for CONS. Blood work from today: Hb 85, Plts 60, WBC 2, ALT 84, AST 64, fibrinogen 0.8g/L, CRP 240, ESR 10, ferritin 18,000. She now reports new abdominal pain and her parents note some confusion at the bedside. On exam, she is febrile 39.8, HR 100, BP 90/60 and is jaundiced with mild hepatosplenomegaly.

Which of the following is the best management?

- A. IV amphotericin B
- B. IV high dose methylprednisolone
- C. PLEX
- D. Rituximab
- E. Cyclophosphamide

B. IV high dose methylprednisolone.

This patient has suspected Still's disease now complicated by macrophage activation syndrome on basis of fever, rash, cytopenias, hyperferritinemia, hepatosplenomegaly. High dose IV steroids are considered first line. Other potential options include CYC or Rituximab, but these are not first line for MAS. There is no indication for anti-fungal treatment at this time.

MCQ Bonus #2 2026

A 27-year-old male with ankylosing spondylitis presents to the outpatient clinic with persistent axial symptoms with BASDAI of 8 (high disease activity). He has tried Naproxen for 4 weeks with minimal improvement in his symptoms, prior to that he was on Celebrex, which he was not able to tolerate. He reports two-week history of right eye pain, erythema and photophobia and was seen by ophthalmology and diagnosed with uveitis. He has been managing on topical steroid drops, but as per ophthalmology there has been no improvement thus far and they feel he requires escalation of treatment.

Which of the following is the most definitive management strategy?

- A. Prednisone 20mg OD with taper 5mg/week
- B. Etanercept 50mg sc qweekly
- C. Adalimumab 40mg sc q2 weeks
- D. Tofacitinib (JAKi) 5mg po BID

MCQ Bonus #2 2026

A 27-year-old male with chronic low back pain. He has been on Celebrex for 6 months. He has erythema at the injection sites. He is managing his pain with Celebrex and they feel better.

Answer: C

For management of **active** axial spondyloarthritis, 1st line management is: NSAIDs then if inadequate response consideration of advanced therapies such as anti-TNF, IL-17 or JAKi.

- TNFi is preferred over JAKi
- IL-17 is preferred over JAKi unless concomitant/suspected IBD

l back that he eye pain, as been thus far

Which of the

- A. Prednisone
B. Etanercept
C. Adalimumab
D. Tofacitinib

His disease is complicated by uveitis that requires immunosuppression – **monoclonal TNFi** (ADA, INF) are preferred for management of concurrent uveitis or IBD. Etanercept can in fact worsen uveitis. Steroids can be considered for inflammatory eye disease, but will not be indicated for his axial disease.

MCQ Bonus #3 2026

A 67-year-old man with a history of gout and stage 4 chronic kidney disease (eGFR 20 mL/min/1.73m²) presents 3 weeks after starting allopurinol for urate lowering. He reports fever, malaise, and a new widespread rash. On exam, you note diffuse erythematous rash with desquamation, facial edema, and cervical lymphadenopathy. Investigations reveal the following: Hb 110, WBC 5, Eos 2, Plts 180, Cr 350 (baseline 280), UA proteinuria and microscopic hematuria.

Which of the following is the most appropriate next step in management?

- A. Continue allopurinol but add prednisone to control inflammation
- B. Stop allopurinol immediately and start high-dose systemic corticosteroids
- C. Switch to febuxostat as an alternative urate-lowering therapy
- D. Add hemodialysis for AKI and assistance with urate clearance acutely

MCQ Bonus #3 2026

- A 67-year-old male with a history of gout and chronic kidney disease (eGFR 20 mL/min/1.73 m²) is on allopurinol 300 mg daily. He reports a new rash and fever. Physical exam shows diffuse erythema and lymphadenopathy. Labs show WBC 18,000/mm³, Cr 1.8 mg/dL, and Plts 150,000/mm³.
- Answer: B
- This patient has allopurinol hypersensitivity syndrome. Individuals of certain ethnic backgrounds (Han Chinese, Thai, Korean) are at higher risk of this, but can happen to others as well (risks higher initial dose of allopurinol and renal dysfunction). AHS can lead to a severe drug reaction characterized by rash, fever, eosinophilia, hepatitis, nephritis, and potentially life-threatening systemic involvement (a DRESS-like syndrome). Must STOP allopurinol and cannot be resumed ever. Glucocorticoids are often used to treat especially if severe organ involvement. A is incorrect as must stop allopurinol. Febuxostat can be used for longer term management but is inappropriate in this context. IHD will not help with AHS.
- Which of the following is the most appropriate next step in management?
- A. Continue allopurinol and add glucocorticoids
 - B. Stop allopurinol and add glucocorticoids
 - C. Switch to febuxostat as an alternative urate-lowering therapy
 - D. Add hemodialysis for AKI and assistance with urate clearance acutely

MCQ Bonus #4 2026

A 33-year-old patient with Sjogren's syndrome on the basis of positive lip biopsy with focus score >3, oral and ocular sicca symptoms and intermittent parotid gland swelling along with positive serology for ANA 1:320, anti-SSA, and RF comes to see you in clinic as she is hoping to become pregnant. She is worried about how Sjogren's may affect her pregnancy. She is currently not on any medication.

What do you counsel her on?

- A. Recommend initiation of hydroxychloroquine
- B. Recommend initiation of low dose prednisone
- C. Recommend ASA 162mg QHS prior to 16 weeks
- D. Recommend she defer pregnancy for now

MCQ Bonus #4 2026

A 3 **Answer A.**

>3, Hydroxychloroquine is recommended in this case given the presence of anti-SSA ser antibodies. Anti-SSA may passively transfer across the placenta and is pre associated with low risk (1-3%) of congenital heart block. There is also low risk any of neonatal lupus, a transient issue related to persistent circulation of maternal Wh antibodies in the newborn that can be associated with cytopenias, transaminitis and a rash. Hydroxychloroquine appears to mitigate this risk as per the PATCH A. study – reduced rate of recurrent congenital heart block by ~50%. This only B. needs to be continued during pregnancy as there is no concerns regarding C. transfer during breastfeeding. There is no indication based on the stem for use D. of prednisone. ASA in pregnancy may be considered by OB in patients with autoimmune diseases (usually SLE given elevated risk of pre-eclampsia), but does not need to be continued post partum.

MCQ Bonus #5 2026

A 69 year old female with rheumatoid arthritis presently managed on methotrexate 15mg weekly and leflunomide 20mg daily presents with worsening diffuse joint pain. At her last visit, you instructed her to taper off her longstanding prednisone, and she has been off this medication now for the past 6 weeks. She feels her pain has worsened since discontinuation of prednisone. She is convinced this is a flare of her rheumatoid arthritis and would like to resume prednisone as her symptoms were previously well controlled. On assessment, you note some chronic deformities of the hands with ulnar drift and MCP subluxation. She is diffusely tender to palpation of all joints in the hands but also soft tissues as well. On point of care ultrasound, you do not note any evidence of joint synovitis. Her most recent monitoring blood work showed the following: Hb 120, WBC 5, Plts 210, Cr 78 with eGFR >120, ESR 5, CRP 3.5

What do you recommend at this time?

- A. Increase her dose of methotrexate
- B. Resume low dose prednisone
- C. Short course of hydromorphone PRN
- D. Initiate duloxetine
- E. Intra-articular injections

MCQ Bonus #5 2026

Answer D.

There is no clear evidence of rheumatoid arthritis disease activity at present time with no clinical or ultrasound evidence of synovitis and no systemic markers of inflammation (normal CRP/ESR). There is no clear indication to increase her dose of methotrexate or resume her prednisone. As per Choosing Wisely, it is not recommended to prescribe opioids for chronic rheumatic disease before optimizing non-opioid strategies, which has yet to occur in this case. While IA GC could be of benefit to some joints, her pain is diffuse and has non-articular areas, which may be more suggestive of a central pain sensitization (i.e. fibromyalgia). Duloxetine has evidence based for management of fibromyalgia. Of note, duloxetine has also been approved for use in OA, particularly multi-joint and knee OA and may have added benefit for patients with concomitant depression and/or pain amplification syndromes.

MCQ Bonus #6 2026

A 37 year old female presents with 2 month history of fatigue, arthralgias, and dry cough. She has erythematous tender nodules over her shins and reports pain and swelling in both ankles. She denies fevers, weight loss, or visual changes. On examination, she has bilateral ankle swelling and erythema nodosum. Chest auscultation is clear. Laboratory findings show: Hb 118, WBC 6.4, Plts 210, CRP 18, ESR 62, Cr 78 with eGFR >120, serum calcium 2.75mmol/L, normal serum ACE, ANA 1:80, RF negative, Quantiferon negative. A chest x-ray shows bilateral hilar lymphadenopathy.

What do you recommend at this time?

- A. Start oral prednisone 40mg
- B. Start methotrexate 25mg subcutaneously
- C. Start NSAIDs
- D. Start hydroxychloroquine 400mg po OD

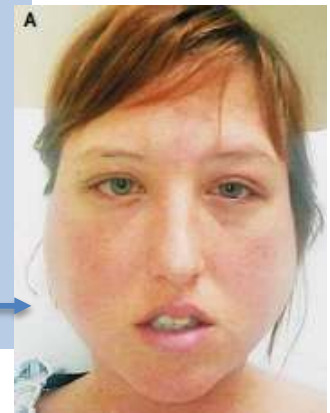
MCQ Bonus #6 2026

Answer C.

This patient's presentation is classic for Löfgren's syndrome — a triad of erythema nodosum, bilateral hilar lymphadenopathy, and ankle arthritis — representing an acute and self-limited form of sarcoidosis usually with an excellent prognosis. Most cases resolve spontaneously but you can consider NSAIDs for symptomatic relief of her joint manifestations. Steroids can be considered if refractory to NSAIDs but are generally reserved for major organ involvement. Methotrexate/HCQ can be used as a steroid sparing agent if chronic or steroid refractory disease, not as first line.

Another rheumatology mimic is Heerfordt Syndrome – can get a bilateral parotiditis which can be mistaken for Sjogren's syndrome. Other features of Heerfordt include uveitis, fever and facial nerve palsy. Seen here

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A Vasculitis Question (that we get asked about every year)

A 24-year-old woman is evaluated in the ER for altered mental status, fatigue, and fever for 4 days. Her husband has noticed that over the last two years she has had tightening and shiny skin, sharp discolouration of her fingertips when exposed to the cold, difficulty opening her mouth fully, and severe GERD. Two weeks ago, she was evaluated in the ER for an asthma flare and started on prednisone 40mg/d for 10 days. At that time, her BP was 110/70 mmHg, and her CBC, renal function and urinalysis were normal. Additional medications are inhaled advair, omeprazole and ventolin prn. On exam, her temperature is 38.3, BP 160/90, HR 102 and RR 24. There is skin thickening over the face, chest, arms hands and legs with evidence of telangectasias. There is 1+ edema of the lower extremities bilaterally with a petechial rash. Blood work: Hb 98 Plt 100, BUN 32, Creat 400, Peripheral smear: schistocytes, Urinalysis: 2+ protein, 3+ blood, no casts

What treatment is the most appropriate for the patient?

- a) Captopril
- b) PLEX + IV pulse steroids
- c) Rituximab
- d) IVIG

That Vasculitis Question

That Vasculitis Question! PLEX + Steroids. Presentation is most consistent with **Thrombotic Thrombocytopenic Purpura** (TTP) given fever, altered mental status, hypertension, petechial rash, hemolytic anemia, and thrombocytopenia. Scleroderma renal crisis is also a consideration given patient has likely systemic sclerosis and was given steroids 2 weeks prior however less likely in setting of fever and significant thrombocytopenia. Treatment of TTP = **PLEX and steroids initially**, rituximab is often added once TTP is confirmed.